

Principles of Statistical Machine Learning

On the Nine Wheels of Statistical Machine Learning

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The Hymn to Mathematics

The Foundation Upon Which We Build

"Mathematics, rightly viewed, possesses not only truth, but supreme beauty cold and austere, like that of sculpture, without appeal to any part of our weaker nature, without the gorgeous trappings of painting or music, yet sublimely pure, and capable of a stern perfection such as only the greatest art can show. The true spirit of delight, the exaltation, the sense of being more than Man, which is the touchstone of the highest excellence, is to be found in mathematics as surely as in poetry."

- BERTRAND RUSSELL, *Study of Mathematics*

The Unreasonable Effectiveness of Mathematics

Wigner's Profound Observation

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THE UNREASONABLE EFFECTIVENESS OF MATHEMATICS IN THE NATURAL SCIENCES

Eugene Wigner

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There is a story about two friends, who were classmates in high school, talking about their jobs. One of them became a statistician and was working on population trends. He showed a reprint to his former classmate. The reprint started, as usual, with the Gaussian distribution and the statistician explained to his former classmate the meaning of the symbols for the actual population, for the

"The miracle of the appropriateness of the language of mathematics for the formulation of the laws of physics is a wonderful gift which we neither understand nor deserve."

The Journey Ahead: The Nine Wheels

An Interlocking System for Statistical Learning

Wheel	Core Question	Key Concept
1.	Problem	Task Definition & Motivation $f : \mathcal{X} \rightarrow \mathcal{Y}$.
2.	Data	Understanding Data structure $\mathcal{D}_n$.
3.	Model	Hypothesis Space $\mathcal{H} \subset \mathcal{Y}^{\mathcal{X}}$.
4.	Principle of Learning	Risk Minimization $R(f) = \mathbb{E}[\mathcal{L}(y, f(\mathbf{x}))]$
5.	Algorithm	Optimization $\mathcal{A}(\mathcal{D}_n, \mathcal{H}, \mathcal{L}) \mapsto \arg \min_{f \in \mathcal{H}} \{\hat{R}(f)\}$
6.	Refinement	Regularization $\arg \min_{f \in \mathcal{H}} \{\hat{R}(f) + \lambda \Omega(f)\}$
7.	Validation	Generalization $\hat{R}_{\text{test}}(f) := \frac{1}{ \mathcal{D}_{\text{test}} } \sum_{i=1}^n \mathbb{1}_{\{obs_i \in \mathcal{D}_{\text{test}}\}} \mathcal{L}(y_i, f(\mathbf{x}_i))$
8.	Theory and Inference	Guarantees & Bounds $\mathbb{P}\left(R(f) - \hat{R}_n(f) \geq \epsilon\right) \leq 2 \exp(-2n\epsilon^2)$
9.	Deployment	Impact & Value Trust = $\mathbb{1}_{\{\text{Model is accurate}\}} \cdot \mathbb{1}_{\{\text{Model is interpretable}\}}$

The machine only learns if all the nine wheels are turning together.

"The formulation of a problem is often more essential than its solution."

- Albert Einstein

Wheel 1: Problem Formulation & Intuition

The “Why” Before the “How”

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- **The “Why”:** What is the real-world pain point? Forecasting? Diagnosis? Discovery?
- **Defining the Task:**
 - **Supervised (Prediction):** Learn $f : \mathcal{X} \rightarrow \mathcal{Y}$.
 - **Unsupervised (Patterns):** Find structure in $\mathcal{X}$.
 - **Reinforcement (Sequential):** Learn policy: $\text{State} \times \text{Action} \rightarrow \text{Reward}$.
- **Success Metrics:** Define *measurable outcomes* **before** building.

$$\mathbb{1}_{\{\text{Model solves business problem}\}} = 1$$

Wheel 1 in Action: From Problem to Formulation

Bridging the Business-Technical Gap

The Business Problem (The “Why”)

E-commerce Site: “Our customer churn is too high. We want to identify customers who are about to leave so we can intervene with a retention offer.”

⇓ **Translates to** ⇓

The ML Task Formulation (The “What”)

- **Task Type: Supervised Learning** (Binary Classification)
- **Input ($\mathcal{X}$):** Customer features (purchase history, login frequency, session duration, etc.)
- **Target ($\mathcal{Y}$):** ‘WillChurnWithin30Days’ = {‘Yes’, ‘No’}
- **Objective Function:** Learn $f : \text{CustomerData} \rightarrow \{\text{Yes}, \text{No}\}$

The Key Takeaway

A well-formulated problem directly **mirrors** and **constrains** the business objective. The ML task is a precise, data-driven reflection of the original “Why”.

Wheel 1: Defining Success & Metrics

How will you know if it's working?

- **The Wrong Question:** “What’s the model’s accuracy?”
- **The Right Question:** “Does this model solve our business problem effectively and efficiently?”

Choosing the Right Metric

Business Goal	ML Task	Key Metric(s)
Diagnose rare disease	Classification	Recall , Specificity
Recommend products	Ranking	Precision@K , NDCG
Forecast revenue	Regression	MAPE , RMSE
Maximize ad revenue	Reinforcement Learning	Total Cumulative Reward

The Metric is a Policy Tool

The metric you choose **directs** the model’s behavior. Optimizing for recall tells the model: “It’s costlier to miss a positive case (a churning user, a disease) than to have a false alarm.”

Wheel 2: The Dataset $\mathcal{D}_n$

Your Raw Material

"Data is the new oil." - Clive Humby
(... but crude oil must be refined to be useful.)

$$\mathcal{D}_n = \{(\mathbf{x}_i, y_i)\}_{i=1}^n, \quad \mathbf{x}_i \in \mathcal{X} \subseteq \mathbb{R}^p, \quad y_i \in \mathcal{Y}$$

- **The Fundamental Assumption:** $\mathcal{D}_n$ is an **i.i.d.** sample from $p(\mathbf{x}, y)$.
- **Data > Algorithm:** Garbage In = Garbage Out.

$$\mathbb{1}_{\{\text{Data is clean}\}} \approx 0$$

is the primary cause of failure.

- **The Curse of Dimensionality:** As $p \uparrow$, data becomes exponentially sparse. Volume grows, insight hides.

$$\mathcal{D}_n := \left\{ (\mathbf{x}_i, y_i) \stackrel{iid}{\sim} p(\mathbf{x}, y), \mathbf{x}_i \in \mathcal{X}, y_i \in \mathcal{Y}, i = 1, \dots, n \right\}$$

Wheel 2: The Dataset $\mathcal{D}_n$

The Raw Material: Beyond “The New Oil”

“Data is not the new oil. It’s the new soil.” - David McCandless
(... a living, fertile foundation for growth.)

The IBM 5 Vs of Data

Volume n (samples) $\times$ p (features). The **scale**.

Velocity The rate of data generation and change. Impacts **streaming** learning.

Variety Structured, unstructured, text, images. Defines the **input space** $\mathcal{X}$.

Veracity Noise, missingness, biases. The **quality** $\mathbb{1}_{\{\text{Data is reliable}\}}$.

Value The potential insight. The **motivation** for Wheels 1–9.

Some Fundamental Definitions/Concepts

$$\mathcal{D}_n = \{(\mathbf{x}_i, y_i)\}_{i=1}^n \stackrel{\text{i.i.d.}}{\sim} p(\mathbf{x}, y)$$

where:

- n is the sample size; $p := \dim(\mathcal{X})$.
- $\mathbf{x}_i \in \mathcal{X} \subseteq \mathbb{R}^p$: **Feature Vector**
- $y_i \in \mathcal{Y}$: **Response/Target**
- $p(\mathbf{x}, y)$: **Joint Distribution or Mother**
- $\mathcal{X} \times \mathcal{Y}$ is the population.
- $\mathcal{D}_n \subset \mathcal{X} \times \mathcal{Y}$.

The **i.i.d.** assumption is the bedrock of most statistical learning theory.

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Wheel 2: The Linear Algebra of Data

Matrices, Correlations, and Geometry

The entire dataset can be represented as an $n \times p$ data matrix $\mathbf{X}$ and a response vector $\mathbf{y}$:

$$\mathbf{X} = \begin{pmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,p} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n,1} & x_{n,2} & \cdots & x_{n,p} \end{pmatrix} \in \mathbb{R}^{n \times p}, \quad \mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \in \mathbb{R}^n$$

Each row is an **observation vector** $\mathbf{x}_i^T$. Each column is a **feature vector** $\mathbf{x}_{(:,j)} \in \mathbb{R}^n$.

Sample Mean Vector

$$\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i = \frac{1}{n} \mathbf{X}^T \mathbf{1}_n$$

Sample Covariance Matrix

$$\mathbf{S} = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T = \frac{1}{n-1} \tilde{\mathbf{X}}^T \tilde{\mathbf{X}}$$

where $\tilde{\mathbf{X}}$ is the centered data matrix.

The **correlation matrix** $\mathbf{R}$ has elements $r_{jk} = \frac{s_{jk}}{\sqrt{s_{jj}s_{kk}}}$, quantifying linear relationships between features j and k .

Wheel 2: Attitude Dataset Preview

Understanding the Data Structure

	rating	complaints	privileges	learning	raises	critical	advance
1	43	51	30	39	61	92	45
2	63	64	51	54	63	73	47
3	71	70	68	69	76	86	48
4	61	63	45	47	54	84	35
5	81	78	56	66	71	83	47
6	43	55	49	44	54	49	34

Dataset Characteristics

- **Source:** R built-in dataset `attitude`
- **Dimensions:** 30 observations $\times$ 7 numeric features
- **Context:** Survey from department employees rating various aspects
- **Target:** rating - overall rating score (dependent variable)
- **Features:** Specific aspect ratings (independent variables)

Wheel 2: Pima Indians Diabetes Dataset

Binary Classification Task Structure

diabetes	pregnant	glucose	pressure	triceps	insulin	mass	pedigree	age
1 pos	6	148	72	35	0	33.6	0.627	50
2 neg	1	85	66	29	0	26.6	0.351	31
3 pos	8	183	64	0	0	23.3	0.672	32
4 neg	1	89	66	23	94	28.1	0.167	21
5 pos	0	137	40	35	168	43.1	2.288	33
6 neg	5	116	74	0	0	25.6	0.201	30

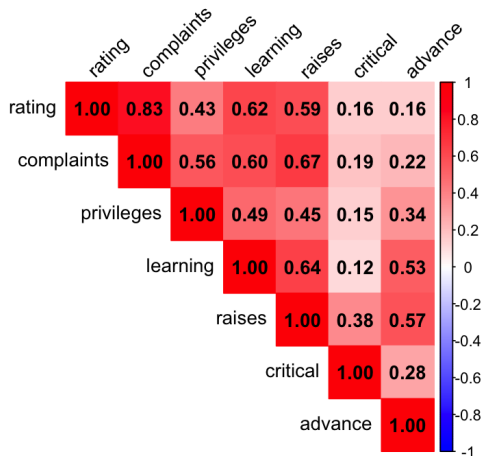
Dataset Characteristics

- **Source:** `library(mlbench), data(PimaIndiansDiabetes)`
- **Dimensions:** 768 observations $\times$ 9 variables (8 features + 1 target)
- **Task:** Binary classification (diabetes: pos/neg)
- **Features:** Medical measurements and demographic information
- **Note:** Zero values in some medical measurements represent missing data

Wheel 2: Visualizing Data Structure

Correlation Matrix Heatmap

Correlation Plot of Attitude Dataset

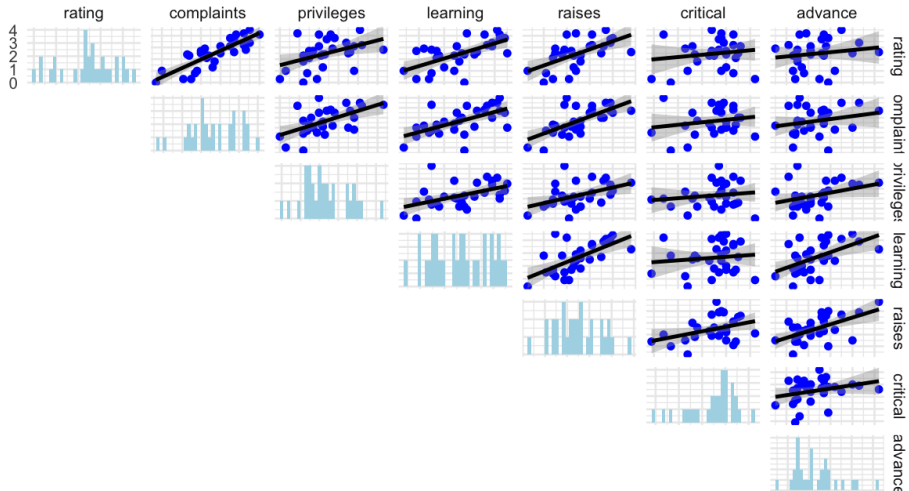


Correlation reveals linear relationships between features

Wheel 2: Pairwise Scatterplots for Regression

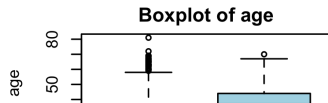
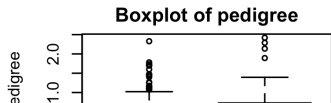
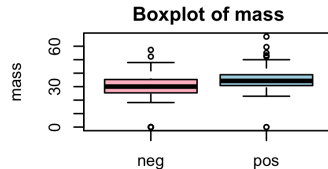
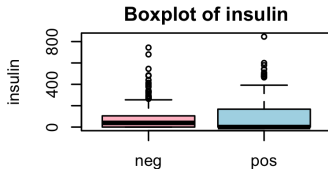
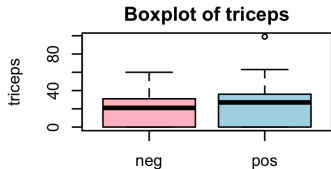
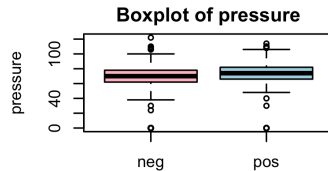
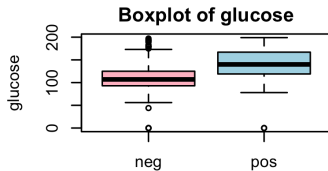
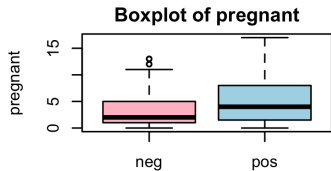
Visualizing Multivariate Relationships

Upper Triangular Matrix of Pairwise Scatterplots with Smooth Regression Estimates



Wheel 2: Comparative Boxplots for Classification

Assessing Feature Importance Across Classes

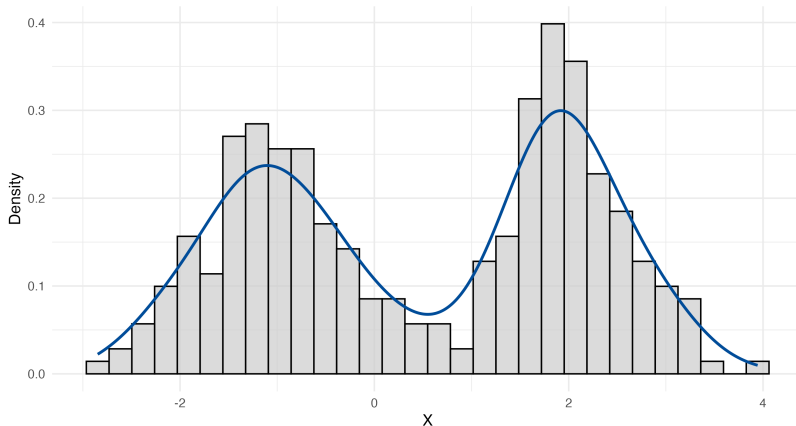


Wheel 2: Nonparametric Density Estimation

Kernel Density Estimation for Data Understanding

Kernel Density Estimation Example

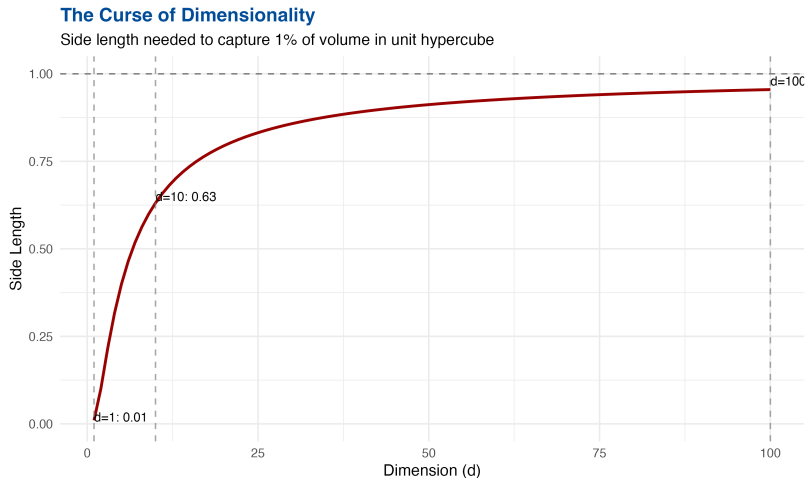
Histogram with Gaussian KDE overlay



- **Kernel Density Estimation (KDE)** reveals underlying data distribution

Wheel 2: The Curse of Dimensionality

Data Sparsity in High Dimensions



- **Exponential data sparsity** as dimensions increase

Wheel 3: The Hypothesis Space $\mathcal{H}$

Your Toolkit of Candidate Functions

"All models are wrong, but some are useful." - George E. P. Box

The set $\mathcal{H}$ restricts our search for f .

- **The Bias-Variance Tradeoff is born here:**

- **High-Bias** $\mathcal{H}$ (e.g., Linear): Simple, stable, may underfit.

$$\mathcal{H}_{\text{lin}} = \{f(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b \mid \mathbf{w} \in \mathbb{R}^p, b \in \mathbb{R}\}$$

- **High-Variance** $\mathcal{H}$ (e.g., DNNs): Flexible, powerful, may overfit.

$$\mathcal{H}_{\text{DNN}} = \{f(\mathbf{x}) = \mathbf{W}_L \sigma(\dots \sigma(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) \dots) + \mathbf{b}_L\}$$

- **Choosing** $\mathcal{H}$ is choosing your prior belief about the data's structure.

Wheel 3: Rich Landscape of Function Spaces

From Simple to Complex Hypothesis Spaces

Linear Models

$$\mathcal{H}_{\text{lin}} = \{f(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b\}$$

- Interpretable, convex optimization
- Strong assumptions about linearity

Polynomial Models

$$\mathcal{H}_{\text{poly}} = \{f(\mathbf{x}) = \sum_{|\alpha| \leq d} w_{\alpha} \mathbf{x}^{\alpha}\}$$

- Universal approximators (Weierstrass)
- Prone to overfitting, numerical issues

Radial Basis Functions

$$\mathcal{H}_{\text{RBF}} = \{f(\mathbf{x}) = \sum_{i=1}^m w_i \phi(\|\mathbf{x} - \mathbf{c}_i\|)\}$$

- Local approximators, mesh-free methods
- Choice of centers $\mathbf{c}_i$ critical

Neural Networks

$$\mathcal{H}_{\text{NN}} = \{f(\mathbf{x}) = \mathbf{W}_L \sigma(\mathbf{W}_{L-1} \sigma(\cdots))\}$$

- Universal approximators (Cybenko)
- Black box, requires careful regularization

Wheel 3: The Wisdom of the Choice

Navigating the Bias-Variance Tradeoff

"Everything should be made as simple as possible, but not simpler."

Albert Einstein

The choice of $\mathcal{H}$ is the primary determinant of the **Bias-Variance Decomposition** of the expected prediction error:

$$\mathbb{E} \left[(y - \hat{f}(\mathbf{x}))^2 \right] = \underbrace{\left(\mathbb{E}[\hat{f}(\mathbf{x})] - f(\mathbf{x}) \right)^2}_{\text{Bias}^2} + \underbrace{\mathbb{E} \left[(\hat{f}(\mathbf{x}) - \mathbb{E}[\hat{f}(\mathbf{x})])^2 \right]}_{\text{Variance}} + \underbrace{\sigma_\epsilon^2}_{\text{Irreducible Error}}$$

Simple $\mathcal{H}$ (e.g., Linear)

- **High Bias** $\mathbb{1}_{\{\text{Model is too simple}\}} = 1$
- **Low Variance**
- **Interpretable**
- Risk: **Underfitting**
 $\mathbb{1}_{\{\text{Captures pattern}\}} \approx 0$

Complex $\mathcal{H}$ (e.g., DNNs)

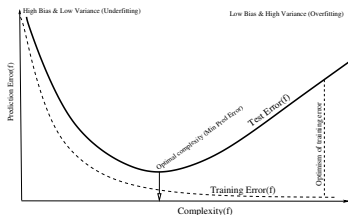
- **Low Bias**
- **High Variance** $\mathbb{1}_{\{\text{Model is too complex}\}} = 1$
- **Black Box**
- Risk: **Overfitting**
 $\mathbb{1}_{\{\text{Captures noise}\}} \approx 1$

The fundamental task is to find the $\mathcal{H}$ whose complexity is **justified by the data**, navigating the U-shaped curve of total error. This is the essence of **Occam's Razor** for machine learning.

Wheel 3: The Bias-Variance Tradeoff

The Fundamental Dilemma of Learning

"For every complex problem there is an answer that is clear, simple, and wrong."
- H.L. Mencken

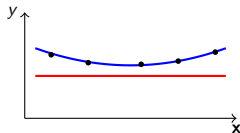


- The V-shaped curve of **Total Error** reveals the central dilemma: **Simplicity** (High Bias) vs. **Complexity** (High Variance).
- The **Irreducible Error** (σ_{ϵ}^2) is the unavoidable noise in the data, our fundamental limit.
- The goal is not to minimize bias *or* variance, but to find the **harmonious minimum** where their sum is minimized. This is the wisdom of model selection.
- This optimum is the complexity **justified by the data**—the model that learns the signal without worshipping the noise.

Wheel 3: Bias-Variance Scenarios

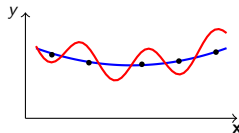
Understanding Different Learning Outcomes

High Bias, Low Variance



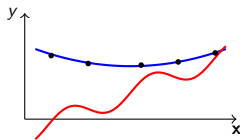
Underfitting: Model too simple

Low Bias, High Variance



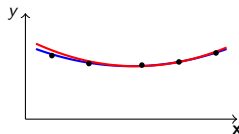
Overfitting: Model too complex

High Bias, High Variance



Worst case: Wrong and unstable

Low Bias, Low Variance



Ideal: Accurate and stable

Mathematical Interpretation:

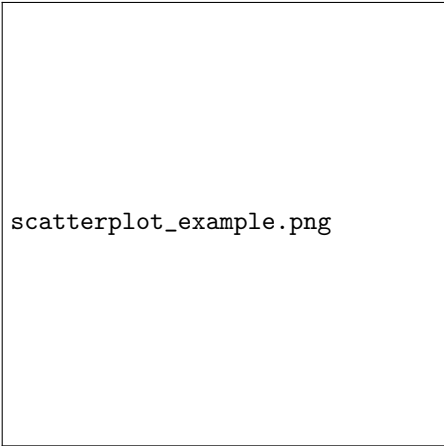
Practical Guidance:

Wheel 3: From Data Visualization to Function Space

From Scatterplots to Mathematical Abstraction

"The purpose of models is not to fit the data but to sharpen the questions." - Samuel Karlin

The Visual Foundation



scatterplot_example.png

The Abstraction Process

- 1 Observe data patterns visually
- 2 Hypothesize mathematical forms
- 3 Define hypothesis space $\mathcal{H}$
- 4 Select $f \in \mathcal{H}$ that captures patterns
- 5 Validate against unseen data

$$\mathcal{H} \subset \mathcal{Y}^{\mathcal{X}}$$

Wheel 3: The Simplest Function Spaces

Foundational Building Blocks

1. Constant Functions

$$\mathcal{H}_{\text{const}} = \{f : \mathcal{X} \rightarrow \mathcal{Y} \mid f(\mathbf{x}) = c, \forall \mathbf{x} \in \mathcal{X}\}$$

where $\mathcal{Y} = \{c\} \subset \mathbb{R}$.

Properties:

- VC dimension: 1
- Bias: High (unless truth is constant)
- Variance: Zero
- Example: $f(\mathbf{x}) = \bar{y}$ (sample mean)

2. Indicator Functions

$$f(\mathbf{x}) = \mathbb{1}_{\{\mathbf{x} \in A\}} = \begin{cases} 1 & \text{if } \mathbf{x} \in A \\ 0 & \text{otherwise} \end{cases}$$

where $A \subseteq \mathcal{X}$ is a measurable set.

3. Piecewise Constant Functions

$$\mathcal{H}_{\text{pwc}} = \left\{ f(\mathbf{x}) = \sum_{l=1}^L c_l \mathbb{1}_{\{\mathbf{x} \in R_l\}} \right\}$$

where $\{R_l\}_{l=1}^L$ partitions $\mathcal{X}$.

Examples:

- Histogram estimators
- Decision stumps
- Binning transformations

Properties:

Complexity $\propto L$ (number of pieces)

Wheel 3: Linear and Affine Function Spaces

The Workhorses of Statistical Modeling

1. Linear Functions

$$\mathcal{H}_{\text{lin}} = \{f(\mathbf{x}) = \mathbf{w}^T \mathbf{x} \mid \mathbf{w} \in \mathbb{R}^p\}$$

Matrix Representation:

$$\mathbf{f} = \mathbf{X}\mathbf{w}, \quad \mathbf{X} \in \mathbb{R}^{n \times p}$$

Geometric Interpretation:

- Hyperplanes in $\mathbb{R}^p$
- Linear subspaces
- Kernel of linear transformations

2. Affine Functions (Linear + Bias)

$$\mathcal{H}_{\text{affine}} = \{f(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b \mid \mathbf{w} \in \mathbb{R}^p, b \in \mathbb{R}\}$$

Properties:

- Universal approximator for affine relationships
- Closed-form solution (normal equations)
- Convex optimization
- Interpretable coefficients

Normal Equations:

$$\hat{\mathbf{w}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

Wheel 3: Polynomial Function Spaces

From Weierstrass to Practical Applications

Polynomial Functions of Degree d :

$$\mathcal{H}_{\text{poly}}^d = \left\{ f(\mathbf{x}) = \sum_{|\alpha| \leq d} w_{\alpha} \mathbf{x}^{\alpha} \right\}$$

where $\alpha = (\alpha_1, \dots, \alpha_p)$ is a multi-index, $|\alpha| = \sum_{i=1}^p \alpha_i$, and $\mathbf{x}^{\alpha} = x_1^{\alpha_1} \cdots x_p^{\alpha_p}$.

Single Variable ($p = 1$):

$$f(x) = w_0 + w_1x + w_2x^2 + \cdots + w_dx^d$$

Dimensions:

$$\dim(\mathcal{H}_{\text{poly}}^d) = \binom{p+d}{d}$$

Weierstrass Approximation Theorem: For any continuous function $g : [a, b] \rightarrow \mathbb{R}$ and $\epsilon > 0$, there exists a polynomial $p(x)$ such that:

$$\sup_{x \in [a, b]} |g(x) - p(x)| < \epsilon$$

Issues:

- Runge's phenomenon (oscillations at edges)
- Numerical instability
- Overfitting with high d

Wheel 3: Trigonometric and Fourier Spaces

Capturing Periodic Patterns

1. Fourier Basis Functions

$$\mathcal{H}_{\text{Fourier}} = \left\{ f(x) = a_0 + \sum_{k=1}^m [a_k \cos(k\omega x) + b_k \sin(k\omega x)] \right\}$$

Properties:

- Orthonormal basis on $[0, 2\pi]$
- Captures periodic patterns
- Discrete Fourier Transform (DFT) connection
- Useful for time series, signal processing

2. Sine/Cosine Functions

$$\mathcal{H}_{\text{trig}} = \{\sin(\omega x), \cos(\omega x), \sin(2\omega x), \cos(2\omega x), \dots\}$$

Fourier Series Representation: For

$$f \in L^2([-\pi, \pi]):$$

$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} [a_k \cos(kx) + b_k \sin(kx)]$$

where:

$$a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(kx) dx, \quad b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(kx) dx$$

Multivariate Extensions:

$$f(\mathbf{x}) = \sum_{\mathbf{k} \in \mathbb{Z}^p} c_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{x}}$$

Wheel 3: Wavelet Function Spaces

Multi-Resolution Analysis

Wavelet Basis Functions:

$$\psi_{j,k}(x) = 2^{j/2} \psi(2^j x - k), \quad j, k \in \mathbb{Z}$$

Wavelet Decomposition:

$$f(x) = \sum_{k \in \mathbb{Z}} c_{j_0,k} \phi_{j_0,k}(x) + \sum_{j \geq j_0} \sum_{k \in \mathbb{Z}} d_{j,k} \psi_{j,k}(x)$$

Common Wavelets:

- Haar wavelet:

$$\psi(x) = \mathbb{1}_{\{0 \leq x < 1/2\}} - \mathbb{1}_{\{1/2 \leq x < 1\}}$$

- Daubechies wavelets
- Mexican hat wavelet
- Morlet wavelet

Properties:

- Localized in both time and frequency
- Multi-resolution analysis (MRA)
- Sparse representations for piecewise smooth functions
- Excellent for compression, denoising

Discrete Wavelet Transform:

$$W_f(j, k) = \langle f, \psi_{j,k} \rangle = \int_{-\infty}^{\infty} f(x) \psi_{j,k}(x) dx$$

Applications:

- Image compression (JPEG2000)

• Signal denoising

Wheel 3: Radial Basis Function (RBF) Spaces

Localized Approximation

RBF Formulation:

$$\mathcal{H}_{\text{RBF}} = \left\{ f(\mathbf{x}) = \sum_{i=1}^m w_i \phi(\|\mathbf{x} - \mathbf{c}_i\|) \right\}$$

Common RBF Kernels:

- Gaussian: $\phi(r) = e^{-(\epsilon r)^2}$
- Multiquadric: $\phi(r) = \sqrt{1 + (\epsilon r)^2}$
- Inverse multiquadric: $\phi(r) = 1/\sqrt{1 + (\epsilon r)^2}$
- Thin plate spline: $\phi(r) = r^2 \log r$

Properties:

- Mesh-free methods
- Local support (depending on ϕ)
- Universal approximators
- Interpolation properties

Center Selection:

- Random subset of data points
- K-means clustering centers
- Optimized via gradient descent

Matrix Form:

$$\mathbf{f} = \Phi \mathbf{w}, \quad \Phi_{ij} = \phi(\|\mathbf{x}_i - \mathbf{c}_j\|)$$

Wheel 3: Spline Function Spaces

Piecewise Polynomial Smoothness

Spline Definition: A spline of degree d with knots $\{\xi_k\}_{k=1}^K$ is a piecewise polynomial of degree d that is C^{d-1} continuous at the knots.

B-spline Basis:

$$B_{i,1}(x) = \mathbb{1}_{\{\xi_i \leq x < \xi_{i+1}\}}$$

$$B_{i,d}(x) = \frac{x - \xi_i}{\xi_{i+d-1} - \xi_i} B_{i,d-1}(x) + \frac{\xi_{i+d} - x}{\xi_{i+d} - \xi_{i+1}} B_{i+1,d-1}(x)$$

Spline Representation:

$$f(x) = \sum_{i=1}^{K+d} \beta_i B_{i,d}(x)$$

Types of Splines:

- Natural splines: Linear beyond boundary knots
- Smoothing splines: Minimize $\sum (y_i - f(x_i))^2 + \lambda \int (f''(x))^2 dx$
- Regression splines: Fixed number of knots
- Penalized splines: Large number of knots with penalty

Multivariate Splines:

- Tensor product splines
- Thin plate splines
- Additive spline models

Wheel 3: Sobolev and Reproducing Kernel Hilbert Spaces

Infinite-Dimensional Function Spaces

Sobolev Spaces $W^{k,p}(\Omega)$:

$$W^{k,p}(\Omega) = \{f \in L^p(\Omega) : D^\alpha f \in L^p(\Omega), \forall |\alpha| \leq k\}$$

Sobolev Norm:

$$\|f\|_{W^{k,p}} = \left(\sum_{|\alpha| \leq k} \|D^\alpha f\|_{L^p}^p \right)^{1/p}$$

Special Case ($p = 2$):

$$H^k(\Omega) = W^{k,2}(\Omega)$$

Hilbert space with inner product:

$$\langle f, g \rangle_{H^k} = \sum_{|\alpha| \leq k} \langle D^\alpha f, D^\alpha g \rangle_{L^2}$$

Reproducing Kernel Hilbert Space (RKHS):

A Hilbert space $\mathcal{H}$ of functions $f : \mathcal{X} \rightarrow \mathbb{R}$ with reproducing kernel $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ such that:

- 1 $\forall x \in \mathcal{X}, K(\cdot, x) \in \mathcal{H}$
- 2 $\forall f \in \mathcal{H}, \langle f, K(\cdot, x) \rangle = f(x)$ (reproducing property)

Representer Theorem: For empirical risk minimization in RKHS:

$$\hat{f} = \arg \min_{f \in \mathcal{H}} \left\{ \frac{1}{n} \sum_{i=1}^n L(y_i, f(x_i)) + \lambda \|f\|_{\mathcal{H}}^2 \right\}$$

has solution:

$$\hat{f}(x) = \sum_{i=1}^n \alpha_i K(x, x_i)$$

Wheel 3: Neural Network Function Spaces

Universal Approximation and Deep Learning

Neural Network Architecture:

$$f(\mathbf{x}) = \mathbf{W}_L \sigma(\mathbf{W}_{L-1} \sigma(\cdots \sigma(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) \cdots) + \mathbf{b}_{L-1}) + \mathbf{b}_L$$

Activation Functions:

- Sigmoid: $\sigma(z) = \frac{1}{1+e^{-z}}$
- ReLU: $\sigma(z) = \max(0, z)$
- Tanh: $\sigma(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$
- Leaky ReLU: $\sigma(z) = \max(\alpha z, z)$
- Softmax: $\sigma(\mathbf{z})_j = \frac{e^{z_j}}{\sum_k e^{z_k}}$

Depth vs Width:

- Shallow networks: Exponential width needed
- Deep networks: Polynomial width sufficient
- Empirical: Depth helps feature learning

Universal Approximation Theorems:

- **Cybenko (1989):** Single hidden layer with sigmoid activation can approximate any continuous function on compact sets
- **Hornik (1991):** Extension to arbitrary activation functions
- **Modern:** Deep networks with ReLU can approximate with fewer parameters

Wheel 3: Decision Tree and Ensemble Spaces

Nonparametric Piecewise Models

Decision Tree:

$$f(\mathbf{x}) = \sum_{l=1}^L c_l \mathbb{1}_{\{\mathbf{x} \in R_l\}}$$

where $\{R_l\}$ form a recursive partition of $\mathcal{X}$.

Splitting Criteria:

- Classification: Gini impurity, entropy
- Regression: Mean squared error

Properties:

- Highly interpretable
- Handles mixed data types
- No feature scaling needed
- Prone to overfitting

Random Forests:

$$f_{\text{RF}}(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^B T_b(\mathbf{x})$$

Properties:

- Bagging + random feature subsets
- Reduces variance
item Out-of-bag error estimate
- Feature importance measures

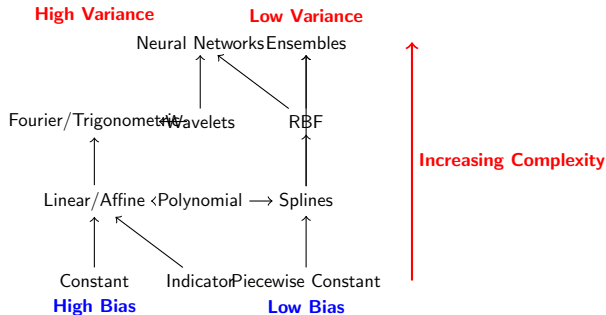
Gradient Boosting:

$$f_m(\mathbf{x}) = f_{m-1}(\mathbf{x}) + \eta \cdot h_m(\mathbf{x})$$

where h_m is a weak learner (typically shallow tree).

Wheel 3: Function Space Hierarchy and Selection

From Simple to Complex: A Continuum



Selection Criteria:

- **Data Size:** $n \ll p$: simple models; $n \gg p$: complex models
- **Problem Domain:** Physics-informed vs. data-driven
- **Interpretability Needs:** Regulatory vs. pure prediction
- **Computational Constraints:** Training vs. inference time

Wheel 3: The Fundamental Reality of Hypothesis Spaces

Constrained Approximation in a Vast Universe of Functions

The Grand Truth: We Always Approximate

$$\mathcal{H} \subsetneq \mathcal{Y}^{\mathcal{X}}$$

No practical hypothesis space $\mathcal{H}$ ever reaches the full power set $\mathcal{Y}^{\mathcal{X}}$ of **all possible mappings** from $\mathcal{X}$ to $\mathcal{Y}$.

The Approximation Error:

$$\text{ApproxError}(\mathcal{H}) = \inf_{f \in \mathcal{H}} R(f) - \inf_{f \in \mathcal{Y}^{\mathcal{X}}} R(f)$$

Interpretation:

- Even the **best** $f \in \mathcal{H}$ may not reach the **optimal** $f^* \in \mathcal{Y}^{\mathcal{X}}$
- This gap is **inevitable**—we trade exactness for tractability
- Choosing $\mathcal{H}$ is choosing **where** to

The Art of Prudent Constraint: We seek $\mathcal{H}$ that is:

- **Rich enough** to capture relevant patterns
- **Constrained enough** to avoid irrelevant regions of $\mathcal{Y}^{\mathcal{X}}$
- **Aligned** with our domain knowledge and task
- **Tractable** for learning and computation

Wheel 4: The Learning Principle — From Space to Selection

The Criterion for Choosing Among Functions

The Fundamental Question

Given $\mathcal{H}$ (or even the mighty $\mathcal{Y}^{\mathcal{X}}$), how do we select the function f that best captures the relationship between $\mathbf{x} \in \mathcal{X}$ and $y \in \mathcal{Y}$?

The Risk Minimization Framework

- 1 Define a **loss function** $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}^+$
- 2 Compute the **risk functional**:

$$R(f) = \mathbb{E}_{(\mathbf{x}, y) \sim p(\mathbf{x}, y)}[\ell(y, f(\mathbf{x}))]$$

- 3 Seek the **optimal learner**:

$$f^* = \arg \min_{f \in \mathcal{F}} R(f)$$

where $\mathcal{F} \subseteq \mathcal{Y}^{\mathcal{X}}$

The Intuition

- ℓ quantifies the **cost of errors**
- $R(f)$ is the **expected cost** over all possible data
- Minimizing $R(f)$ yields the **best predictor** in expectation
- The choice of ℓ encodes our **value judgment** about different types of errors

$$\mathbb{1}_{\{\text{Learning occurs}\}} = \mathbb{1}_{\{\text{Risk is minimized}\}}$$

Wheel 4: Loss Functions for Classification

Quantifying the Cost of Misclassification

The Canonical Loss: 0-1 Loss

$$\ell_{0/1}(y, \hat{y}) = \mathbb{1}_{\{y \neq \hat{y}\}} = \begin{cases} 0 & \text{if } y = \hat{y} \\ 1 & \text{if } y \neq \hat{y} \end{cases}$$

1. Hinge Loss (SVM)

$$\ell_{\text{hinge}}(y, f(\mathbf{x})) = \max(0, 1 - y \cdot f(\mathbf{x}))$$

where $y \in \{-1, +1\}$.

2. Logistic Loss

$$\ell_{\text{logistic}}(y, f(\mathbf{x})) = \log(1 + e^{-yf(\mathbf{x})})$$

3. Exponential Loss (AdaBoost)

$$\ell_{\text{exp}}(y, f(\mathbf{x})) = e^{-yf(\mathbf{x})}$$

4. Cross-Entropy (Log Loss) For K classes with probabilities $p_k = \mathbb{P}(Y = k|\mathbf{x})$:

$$\ell_{\text{CE}}(y, \mathbf{p}) = - \sum_{k=1}^K \mathbb{1}_{\{y=k\}} \log p_k$$

5. Hellinger Loss

$$\ell_{\text{Hellinger}}(y, p) = (\sqrt{y} - \sqrt{p})^2 + (\sqrt{1-y} - \sqrt{1-p})^2$$

Properties:

- **0-1:** Non-convex, non-differentiable, direct
- **Hinge:** Convex, piecewise linear,

Wheel 4: Loss Functions for Regression

Quantifying the Cost of Prediction Errors

1. Squared Error Loss (L_2)

$$\ell_{\text{sq}}(y, \hat{y}) = (y - \hat{y})^2$$

Properties:

- Convex, differentiable
- Sensitive to outliers
- Corresponds to Gaussian noise assumption

2. Absolute Error Loss (L_1)

$$\ell_{\text{abs}}(y, \hat{y}) = |y - \hat{y}|$$

Properties:

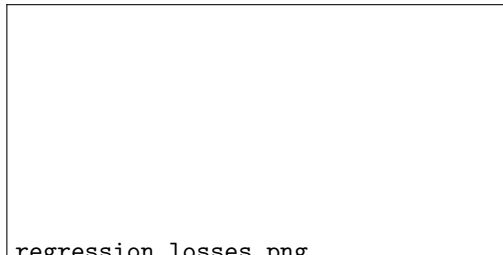
- Convex, not differentiable at 0
- Robust to outliers
- Corresponds to Laplace noise assumption

4. ϵ -Insensitive Loss (SVR)

$$\ell_{\epsilon}(y, \hat{y}) = \max(0, |y - \hat{y}| - \epsilon)$$

5. Quantile Loss For $\tau \in (0, 1)$:

$$\ell_{\tau}(y, \hat{y}) = \begin{cases} \tau(y - \hat{y}) & \text{if } y \geq \hat{y} \\ (1 - \tau)(\hat{y} - y) & \text{if } y < \hat{y} \end{cases}$$



Wheel 4: The Universal Optimal Learner — Bayes' Learner

The Unattainable Gold Standard

Theorem (Bayes Optimal Classifier)

For 0-1 loss $\ell_{0/1}$, the optimal predictor over **all possible functions** $\mathcal{Y}^{\mathcal{X}}$ is:

$$f_{class}^*(\mathbf{x}) = \arg \max_{y \in \mathcal{Y}} \mathbb{P}(Y = y | \mathbf{X} = \mathbf{x})$$

That is: predict the **most probable class** given the features.

Theorem (Bayes Optimal Regressor for L_2 Loss)

For squared error loss ℓ_{sq} , the optimal predictor is:

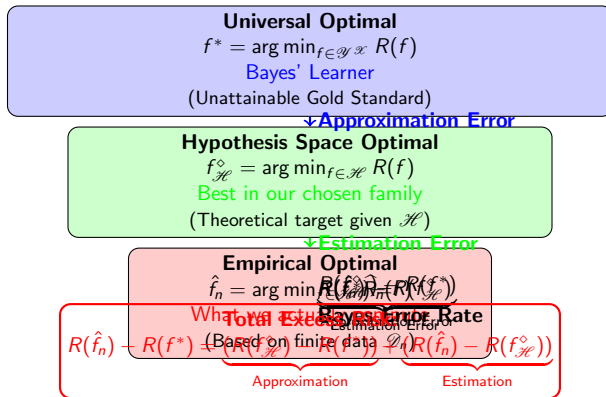
$$f_{L_2}^*(\mathbf{x}) = \mathbb{E}[Y | \mathbf{X} = \mathbf{x}]$$

The **conditional expectation** of Y given $\mathbf{X}$.

Theorem (Bayes Optimal Regressor for L_1 Loss)

Wheel 4: The Fundamental Journey — From Universal to Feasible

The Three Layers of Optimality



The Learning Theorist's Quest

We navigate from the **unattainable universal optimum** f^* down to the **practical empirical estimate** $\hat{f}_n$, always aware of the two fundamental errors we incur:

Wheel 4: Risk Decomposition — The Full Picture

Where All Errors Are Accounted For

Complete Decomposition of Expected Risk:

$$\mathbb{E}[R(\hat{f}_n)] - R(f^*) = \underbrace{\left(\inf_{f \in \mathcal{H}} R(f) - R(f^*) \right)}_{\text{Approximation Error}} + \underbrace{\left(\mathbb{E}[R(\hat{f}_n)] - \inf_{f \in \mathcal{H}} R(f) \right)}_{\text{Estimation Error}}$$

Approximation Error:

$$\mathcal{E}_{\text{app}}(\mathcal{H}) = \inf_{f \in \mathcal{H}} R(f) - R(f^*)$$

- Due to **restricting** to $\mathcal{H} \subset \mathcal{Y}^{\mathcal{X}}$
- **Independent** of sample size n
- Decreases as $\mathcal{H}$ becomes **richer**
- Zero if $f^* \in \mathcal{H}$ (*realizable case*)

Estimation Error:

$$\mathcal{E}_{\text{est}}(n, \mathcal{H}) = \mathbb{E}[R(\hat{f}_n)] - \inf_{f \in \mathcal{H}} R(f)$$

- Due to **finite data** $\mathcal{D}_n$
- Depends on **sample size** n
- Increases with **complexity** of $\mathcal{H}$
- Decreases as $n \rightarrow \infty$ (*consistency*)

The Fundamental Tradeoff

Wheel 4: Proof Sketches for Bayes' Learners

Why These Forms Are Optimal

Theorem (Bayes Optimal Classifier for 0-1 Loss)

For 0-1 loss $\ell_{0/1}(y, \hat{y}) = \mathbb{1}_{\{y \neq \hat{y}\}}$, the optimal predictor is:

$$f^*(\mathbf{x}) = \arg \max_{y \in \mathcal{Y}} \mathbb{P}(Y = y | \mathbf{X} = \mathbf{x})$$

Proof Sketch.

For a fixed $\mathbf{x}$, the conditional risk is:

$$R(f|\mathbf{x}) = \mathbb{E}_{Y|\mathbf{X}=\mathbf{x}}[\mathbb{1}_{\{Y \neq f(\mathbf{x})\}}] = 1 - \mathbb{P}(Y = f(\mathbf{x}) | \mathbf{X} = \mathbf{x})$$

To minimize $R(f|\mathbf{x})$, we maximize $\mathbb{P}(Y = f(\mathbf{x}) | \mathbf{X} = \mathbf{x})$. Thus:

$$f^*(\mathbf{x}) = \arg \max_{y \in \mathcal{Y}} \mathbb{P}(Y = y | \mathbf{X} = \mathbf{x})$$

The overall risk $R(f) = \mathbb{E}_{\mathbf{X}}[R(f|\mathbf{x})]$ is minimized pointwise. □

Wheel 4: Extended Loss Function Zoo

Beyond the Basics

Contrastive Losses For metric learning/siamese networks:

$$\ell_{\text{contrast}}(d) = \begin{cases} \frac{1}{2}d^2 & \text{if } y = 1 \text{ (similar)} \\ \frac{1}{2}\max(0, m - d)^2 & \text{if } y = 0 \text{ (dissimilar)} \end{cases}$$

where $d = \|f(\mathbf{x}_1) - f(\mathbf{x}_2)\|$, m is margin.

Triplet Loss

$$\ell_{\text{triplet}} = \max(0, \|f(a) - f(p)\|^2 - \|f(a) - f(n)\|^2 + m)$$

Anchor a , positive p , negative n .

Wasserstein Distance

$$W_p(P, Q) = \left(\inf_{\gamma \in \Gamma(P, Q)} \int \|x - y\|^p d\gamma(x, y) \right)^{1/p}$$

Optimal transport cost between distributions

Ernest Fokoué, PhD

Focal Loss For class imbalance:

$$\ell_{\text{focal}} = -(1 - p_t)^\gamma \log(p_t)$$

where $p_t = p$ if $y = 1$, else $1 - p$.

Dice Loss For segmentation:

$$\ell_{\text{Dice}} = 1 - \frac{2 \sum p_i g_i + \epsilon}{\sum p_i + \sum g_i + \epsilon}$$

Cosine Similarity Loss

$$\ell_{\text{cos}}(y, \hat{y}) = 1 - \frac{y \cdot \hat{y}}{\|y\| \|\hat{y}\|}$$

Mahalanobis Distance

$$\ell_M(\mathbf{x}, \mu) = \sqrt{(\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu)}$$

On The Nine Wheels of SML

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Wheel 4: Historical Foundations

Giants Whose Shoulders We Stand On

Ronald A. Fisher (1890-1962)

- **Maximum Likelihood Estimation** (1922)
- Analysis of variance (ANOVA)
- Fisher information matrix
- "No causation without manipulation"
- Fisher's exact test

$$\mathcal{L}(\theta; \mathbf{x}) = p(\mathbf{x}|\theta)$$

Abraham Wald (1902-1950)

- **Statistical decision theory**
- Minimax principle
- Sequential analysis
- Wald test
- Complete class theorems

Vladimir Vapnik (1936-present)

- **Statistical Learning Theory**
- VC dimension
- Structural Risk Minimization
- Support Vector Machines
- Empirical Risk Minimization principles

$$R(f) \leq R_n(f) + \sqrt{\frac{h(\log(2n/h) + 1) - \log(\eta/4)}{n}}$$

Thomas Bayes (1701-1761)

- **Bayes' Theorem** (posthumous 1763)

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

- Foundation for Bayesian inference

Wheel 4: The Ubiquitous Maximum Likelihood Principle

Learning Through Probability

The Probabilistic Perspective

Assume data $\mathcal{D}_n = \{(\mathbf{x}_i, y_i)\}$ are i.i.d. from parametric family:

$$p(\mathbf{x}, y; \theta), \quad \theta \in \Theta$$

Likelihood Function:

$$\mathcal{L}(\theta; \mathcal{D}_n) = \prod_{i=1}^n p(\mathbf{x}_i, y_i; \theta)$$

Log-Likelihood:

$$\ell(\theta) = \log \mathcal{L}(\theta; \mathcal{D}_n) = \sum_{i=1}^n \log p(\mathbf{x}_i, y_i; \theta)$$

Definition (Maximum Likelihood Estimator)

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta \in \Theta} \mathcal{L}(\theta; \mathcal{D}_n) = \arg \max_{\theta \in \Theta} \ell(\theta)$$

Wheel 4: MLE Examples

Concrete Applications

Example (Normal Distribution MLE)

For $X_i \stackrel{iid}{\sim} N(\mu, \sigma^2)$:

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

Solving $\frac{\partial \ell}{\partial \mu} = 0$ and $\frac{\partial \ell}{\partial \sigma^2} = 0$ gives:

$$\hat{\mu}_{\text{MLE}} = \bar{x}, \quad \hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

Example (Linear Regression MLE)

Assume $y_i = \mathbf{x}_i^T \beta + \epsilon_i$, $\epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$:

$$\ell(\beta, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{X}\beta\|^2$$

Wheel 4: The Eternal Connection — MLE as ERM

Two Principles, One Essence

Theorem (MLE is ERM with Negative Log-Likelihood Loss)

Maximum Likelihood Estimation is equivalent to Empirical Risk Minimization with loss function:

$$\ell(y, f(\mathbf{x})) = -\log p(y|\mathbf{x}; \theta)$$

Proof.

For i.i.d. data $\{(\mathbf{x}_i, y_i)\}$ with conditional distribution $p(y|\mathbf{x}; \theta)$:

$$\begin{aligned}\hat{\theta}_{\text{MLE}} &= \arg \max_{\theta} \prod_{i=1}^n p(y_i|\mathbf{x}_i; \theta) \\ &= \arg \max_{\theta} \sum_{i=1}^n \log p(y_i|\mathbf{x}_i; \theta) \\ &= \arg \min_{\theta} \sum_{i=1}^n -\log p(y_i|\mathbf{x}_i; \theta)\end{aligned}$$

Wheel 4: The Bayesian Learning Principle

Where Priors Enter the Dance

The Bayesian Philosophy

Prior Beliefs + Observed Data = Updated Beliefs

Bayes' Theorem:

$$p(\theta|\mathcal{D}_n) = \frac{p(\mathcal{D}_n|\theta)p(\theta)}{p(\mathcal{D}_n)} \propto \mathcal{L}(\theta; \mathcal{D}_n) \cdot p(\theta)$$

Components:

- **Prior:** $p(\theta)$ — beliefs before seeing data
- **Likelihood:** $\mathcal{L}(\theta; \mathcal{D}_n)$ — data given parameters
- **Posterior:** $p(\theta|\mathcal{D}_n)$ — updated beliefs
- **Marginal Likelihood:**
 $p(\mathcal{D}_n) = \int p(\mathcal{D}_n|\theta)p(\theta)d\theta$

Bayesian Estimators:

- **Maximum A Posteriori (MAP):**

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(\theta|\mathcal{D}_n)$$

- **Posterior Mean:**

$$\hat{\theta}_{\text{PM}} = \mathbb{E}[\theta|\mathcal{D}_n] = \int \theta p(\theta|\mathcal{D}_n) d\theta$$

- **Posterior Median:** Median of $p(\theta|\mathcal{D}_n)$

Wheel 4: Bayesian Learning in Action

Concrete Examples

Example (Bayesian Linear Regression)

Prior: $\beta \sim N(0, \lambda^{-1}I_p)$

Likelihood: $y_i | \mathbf{x}_i, \beta \sim N(\mathbf{x}_i^T \beta, \sigma^2)$

Posterior:

$$\beta | \mathcal{D}_n \sim N(\hat{\beta}_{\text{Bayes}}, \Sigma_{\text{post}})$$

where

$$\hat{\beta}_{\text{Bayes}} = (\mathbf{X}^T \mathbf{X} + \lambda \sigma^2 I_p)^{-1} \mathbf{X}^T \mathbf{y}$$

$$\Sigma_{\text{post}} = \sigma^2 (\mathbf{X}^T \mathbf{X} + \lambda \sigma^2 I_p)^{-1}$$

Connection to Ridge Regression: MAP estimate = Ridge solution with $\lambda_{\text{ridge}} = \lambda \sigma^2$.

Wheel 4: Bayesian Learning in Action

Concrete Examples

Example (Bayesian Logistic Regression)

Prior: $\beta \sim N(0, \Sigma_0)$

Likelihood: $y_i | \mathbf{x}_i, \beta \sim \text{Bernoulli}(\sigma(\mathbf{x}_i^T \beta))$

Posterior:

$$p(\beta | \mathcal{D}_n) \propto \left[\prod_{i=1}^n \sigma(\mathbf{x}_i^T \beta)^{y_i} (1 - \sigma(\mathbf{x}_i^T \beta))^{1-y_i} \right] \cdot \exp \left(-\frac{1}{2} \beta^T \Sigma_0^{-1} \beta \right)$$

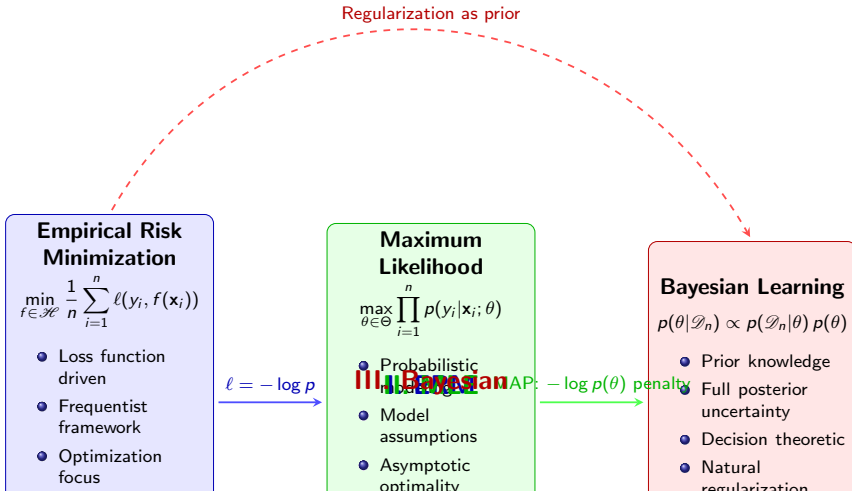
No closed form; use MCMC or variational inference.

Example (Conjugate Priors)

Likelihood	Conjugate Prior	Posterior
Normal (known variance)	Normal	Normal
Binomial	Beta	Beta
Poisson	Gamma	Gamma
Multinomial	Dirichlet	Dirichlet
Normal (both unknown)	Normal-Gamma	Normal-Gamma

Wheel 4: The Three Pillars of Learning

A Unified View of Statistical Learning Principles



Wheel 4: The Three Pillars of Learning Principles

A Unified View

The Grand Synthesis

- **ERM**: Most general framework, minimal assumptions
- **MLE**: Special case of ERM with probabilistic interpretation
- **Bayesian**: Extends MLE with prior knowledge, provides full uncertainty
- **All connected**: Different perspectives on the same fundamental problem

The Learning Principle Summarized

“We seek functions f that minimize expected loss (ERM), often through probabilistic models (MLE), sometimes enhanced with prior knowledge (Bayesian), always aware that our finite data gives us only an approximation of the truth we seek.”

Wheel 4: The Learning Principle — Risk Minimization

From Ideal Theory to Practical Computation

“The goal of learning is generalization, not just fitting the data.” — Vladimir Vapnik

The Ideal (Unattainable):

$$f^* = \arg \min_{f \in \mathcal{Y}^{\mathcal{X}}} R(f)$$

where the **True Risk** is:

$$R(f) = \mathbb{E}_{(\mathbf{x}, y) \sim p(\mathbf{x}, y)} [\ell(y, f(\mathbf{x}))]$$

Problem: We cannot compute $R(f)$ — $p(\mathbf{x}, y)$ is unknown.

The Practical (ERM):

$$\hat{f}_n = \arg \min_{f \in \mathcal{H}} \hat{R}_n(f)$$

where the **Empirical Risk** is:

$$\hat{R}_n(f) = \frac{1}{n} \sum_{i=1}^n \ell(y_i, f(\mathbf{x}_i))$$

Solution: Approximate $R(f)$ with $\hat{R}_n(f)$ using data $\mathcal{D}_n$.

The Fundamental Tradeoff

- **True Risk $R(f)$:** Measures *generalization* — performance on unseen data

Wheel 4: The Learning Principle

Theoretical Risk Minimization

“The goal of learning is generalization, not just fitting the data.” - Vladimir Vapnik

The ultimate, unattainable goal is to find the perfect function from our hypothesis space:

$$f^* = \arg \min_{f \in \mathcal{H}} R(f), \quad \text{where} \quad R(f) = \mathbb{E}_{(\mathbf{x}, y) \sim p(\mathbf{x}, y)} [\ell(f(\mathbf{x}), y)]$$

- **The Problem:** We cannot compute $R(f)$ because $p(\mathbf{x}, y)$ is unknown.
- **The Solution:** We resort to the **Empirical Risk Minimization (ERM)** principle, using our finite sample $\mathcal{D}_n$ as a proxy.

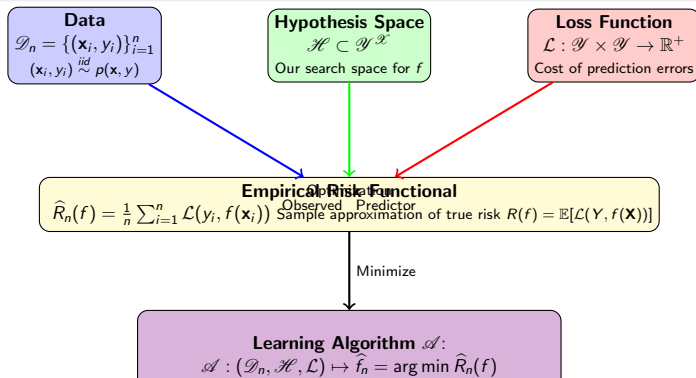
Wheel 5: The Algorithmic Synthesis

Where Theory Becomes Computation

The Algorithmic Trinity

We bring together all elements to create a learning machine:

$$\hat{f}_n = \mathcal{A}(\mathcal{D}_n, \mathcal{H}, \mathcal{L})$$



Wheel 5: Concrete Example I — Polynomial Regression

Ordinary Least Squares in Action

Problem Setup:

- Input: $\mathcal{X} = \mathbb{R}$ (univariate)
- Output: $\mathcal{Y} = \mathbb{R}$ (regression)
- Hypothesis Space: Degree- p polynomials

$$\mathcal{H}_p = \{f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots + \beta_p x^p\}$$

- Parameters: $\beta = (\beta_0, \beta_1, \dots, \beta_p)^T \in \mathbb{R}^{p+1}$
- Loss: Squared error

$$\ell(y, \hat{y}) = (y - \hat{y})^2$$

Empirical Risk:

$$\hat{R}_n(\beta) = \frac{1}{n} \sum_{i=1}^n (y_i - f_\beta(x_i))^2$$

where $f_\beta(x_i) = \beta_0 + \beta_1 x_i + \dots + \beta_p x_i^p$.

Matrix Formulation: Design matrix $\mathbf{X} \in \mathbb{R}^{n \times (p+1)}$:

$$\mathbf{X} = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^p \\ 1 & x_2 & x_2^2 & \cdots & x_2^p \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^p \end{bmatrix}$$

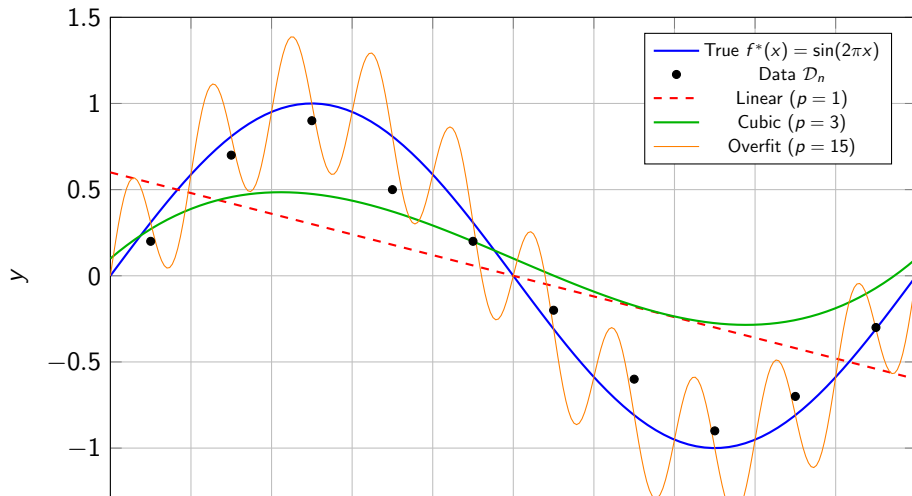
Then:

$$\hat{R}_n(\beta) = \frac{1}{n} \|\mathbf{y} - \mathbf{X}\beta\|^2$$

Wheel 5: Polynomial Regression Visualization

Fitting Different Degrees

Polynomial Regression: Bias-Variance Tradeoff



Wheel 5: Concrete Example II — Logistic Regression

Binary Classification in $\mathbb{R}^p$

Problem Setup:

- Input: $\mathcal{X} = \mathbb{R}^p$ (multivariate)
- Output: $\mathcal{Y} = \{0, 1\}$ (binary classification)
- Hypothesis Space: Linear decision boundary

$$\mathcal{H} = \{f(\mathbf{x}) = \sigma(\mathbf{w}^T \mathbf{x} + b)\}$$

where $\sigma(z) = 1/(1 + e^{-z})$ is sigmoid

- Parameters: $\mathbf{w} \in \mathbb{R}^p$, $b \in \mathbb{R}$

Probabilistic Interpretation:

$$\mathbb{P}(Y = 1 | \mathbf{X} = \mathbf{x}) = \sigma(\mathbf{w}^T \mathbf{x} + b)$$

$$\mathbb{P}(Y = 0 | \mathbf{X} = \mathbf{x}) = 1 - \sigma(\mathbf{w}^T \mathbf{x} + b)$$

Loss Function: Binary Cross-Entropy

$$\ell(y, \hat{p}) = -[y \log \hat{p} + (1 - y) \log(1 - \hat{p})]$$

where $\hat{p} = f(\mathbf{x}) = \sigma(\mathbf{w}^T \mathbf{x} + b)$.

Empirical Risk:

$$\hat{R}_n(\mathbf{w}, b) = -\frac{1}{n} \sum_{i=1}^n [y_i \log \sigma(\mathbf{w}^T \mathbf{x}_i + b) + (1 - y_i) \log(1 - \sigma(\mathbf{w}^T \mathbf{x}_i + b))]$$

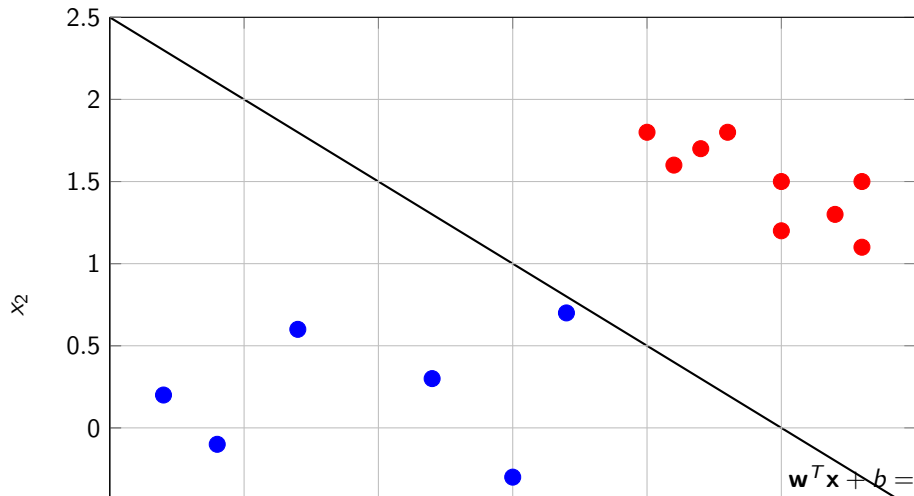
Matrix Form: With $\theta = (b, \mathbf{w}^T)^T$ and $\tilde{\mathbf{X}} = [\mathbf{1}_n, \mathbf{X}]$:

$$\hat{R}_n(\theta) = -\frac{1}{n} [\mathbf{y}^T \log \sigma(\tilde{\mathbf{X}}\theta) + (\mathbf{1}_n - \mathbf{y})^T \log(\mathbf{1}_n - \sigma(\tilde{\mathbf{X}}\theta))]$$

Wheel 5: Logistic Regression Geometry

Decision Boundaries and Probabilities

Logistic Regression: Linear Decision Boundary



Wheel 5: The Famous Excess Risk Decomposition

Understanding Our Limitations

Theorem (Excess Risk Decomposition)

For any learned function $\hat{f}_n$ from hypothesis space $\mathcal{H}$ using data $\mathcal{D}_n$:

$$\underbrace{R(\hat{f}_n) - R(f^*)}_{\text{Excess Risk}} = \underbrace{\left(\inf_{f \in \mathcal{H}} R(f) - R(f^*) \right)}_{\text{Approximation Error}} + \underbrace{\left(R(\hat{f}_n) - \inf_{f \in \mathcal{H}} R(f) \right)}_{\text{Estimation Error}}$$

where:

- $f^* = \arg \min_{f \in \mathcal{Y}^{\mathcal{X}}} R(f)$ — **Bayes optimal** (universal best)
- $f_{\mathcal{H}}^{\diamond} = \arg \min_{f \in \mathcal{H}} R(f)$ — **Best in class** (target given $\mathcal{H}$)
- $\hat{f}_n = \mathcal{A}(\mathcal{D}_n, \mathcal{H}, \mathcal{L})$ — **Empirical minimizer** (what we compute)

Interpretation:

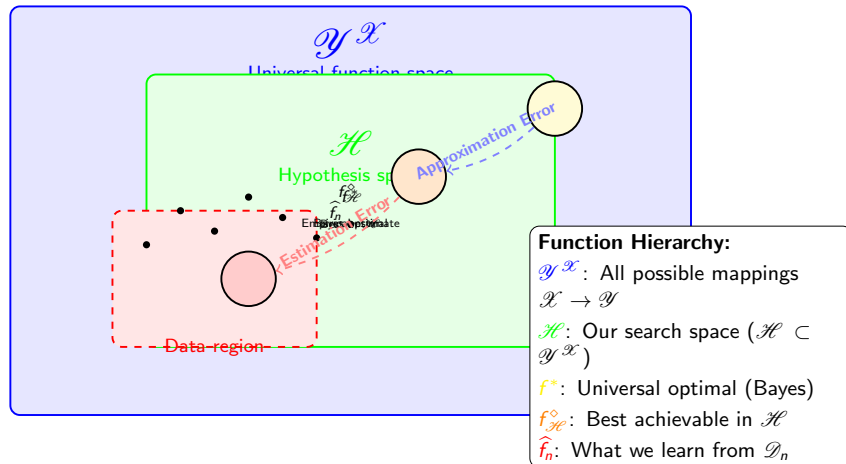
- **Approximation Error:** How good is $\mathcal{H}$?
 - Due to restricting to $\mathcal{H} \subset \mathcal{Y}^{\mathcal{X}}$

Fundamental Tradeoff:

Complexity of $\mathcal{H} \uparrow \Rightarrow \left\{ \begin{array}{l} \text{Approximation Error} \downarrow \\ \text{Estimation Error} \uparrow \end{array} \right.$

Wheel 5: The Three-Layer Hierarchy Visualized

From Universal Optimal to Empirical Estimate

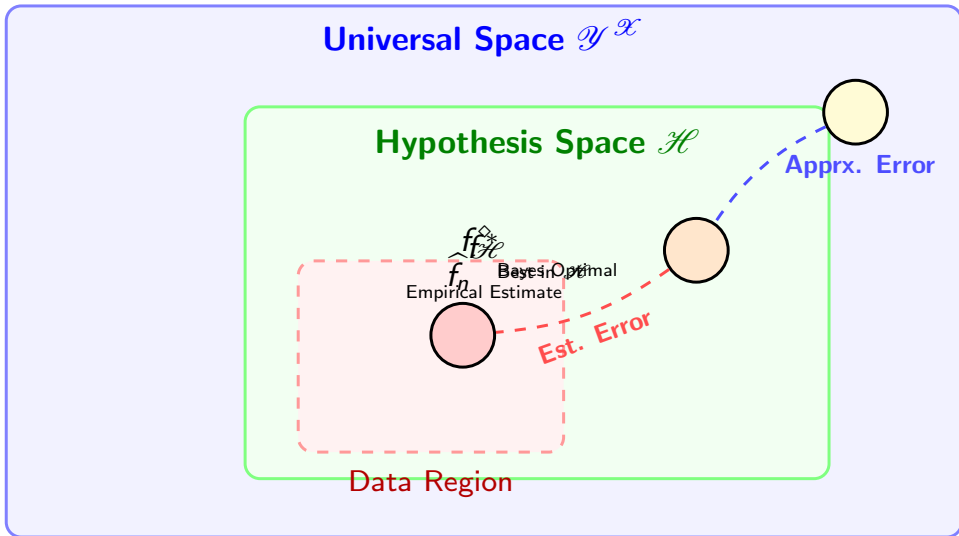


Mathematical Journey:

Practical Interpretation:

Wheel 5: The Three-Layer Hierarchy Visualized

From Universal Optimal to Empirical Estimate



Wheel 5: Decomposition Deep-Dive

Understanding Error Components

Definitions:

- $R(f) = \mathbb{E}[\mathcal{L}(Y, f(X))]$
- $f^* = \arg \min_{f \in \mathcal{Y}} \mathcal{R}(f)$
- $f_{\mathcal{H}}^{\diamond} = \arg \min_{f \in \mathcal{H}} R(f)$
- $\hat{f}_n = \arg \min_{f \in \mathcal{H}} \hat{R}_n(f)$

Key Components:

- **Bayes Error:** Fundamental limit.
- **Excess Risk:** Performance gap.

Practical Interpretation:

① Approximation Error

- How good is $\mathcal{H}$?
- Decreases with richer $\mathcal{H}$.

② Estimation Error

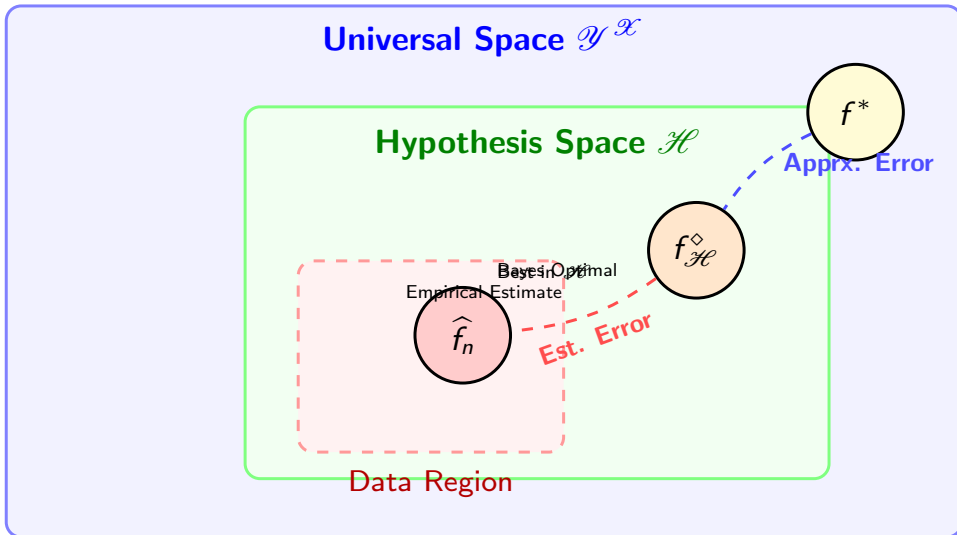
- Cost of finite data n .
- Increases with complex $\mathcal{H}$.

The Learning Algorithm's Challenge

Design an algorithm $\mathcal{A}$ that finds $\hat{f}_n$ close to $f_{\mathcal{H}}^{\diamond}$, while choosing $\mathcal{H}$ that makes $f_{\mathcal{H}}^{\diamond}$ close to f^* , balancing capacity against finite data $\mathcal{D}_n$.

Wheel 5: The Three-Layer Hierarchy Visualized

From Universal Optimal to Empirical Estimate



Wheel 5: The Three-Layer Hierarchy Visualized

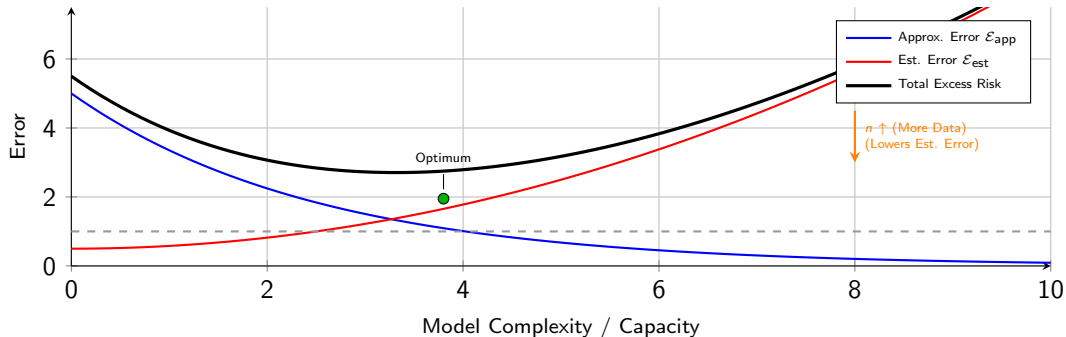
From Universal Optimal to Empirical Estimate

The Algorithm Designer's Challenge

Design $\mathcal{A}$ that efficiently finds $\hat{f}_n$ close to $f_{\mathcal{H}}^{\diamond}$, while choosing $\mathcal{H}$ that makes $f_{\mathcal{H}}^{\diamond}$ close to f^* , all within computational constraints.

Wheel 5: Error Decomposition in Practice

Bias-Variance Trade-off via Complexity and Data



Observations:

- **Simple models:** High approx, low estimation error (Underfitting).
- **Complex models:** Low approx, high estimation error (Overfitting).
- **Bayes error:** The floor.

Practical Guidelines:

- 1 **High training error:** Increase complexity (deeper nets, more features).
- 2 **Large gap:** Reduce complexity or add **regularization**.
- 3 **More data:** Moves the "Optimum" to the right.

Wheel 5: Algorithmic Implementation

Empirical Risk Minimization

We approximate the true objective by minimizing the empirical risk:

$$\hat{f} = \arg \min_{f \in \mathcal{H}} \hat{R}_n(f), \quad \text{where} \quad \hat{R}_n(f) = \frac{1}{n} \sum_{i=1}^n \ell(f(\mathbf{x}_i), y_i)$$

For models parameterized by θ (e.g., $f(\mathbf{x}) = f(\mathbf{x}; \theta)$), we solve:

$$\hat{\theta} = \arg \min_{\theta \in \Theta} J(\theta), \quad J(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(f(\mathbf{x}_i; \theta), y_i)$$

This optimization is almost always solved **iteratively** via gradient-based methods.

Wheel 5: The Optimization Algorithm

Gradient Descent in Practice

- 1: **Given:** Learning rate η , initial parameters $\theta^{(0)}$, tolerance ϵ , max iterations T
- 2: $t \leftarrow 0$
- 3: **repeat**
- 4: Compute gradient: $\mathbf{g}^{(t)} \leftarrow \nabla_{\theta} J(\theta^{(t)})$
- 5: Update parameters: $\theta^{(t+1)} \leftarrow \theta^{(t)} - \eta \cdot \mathbf{g}^{(t)}$
- 6: $t \leftarrow t + 1$
- 7: **until** $\|\mathbf{g}^{(t)}\|_2 < \epsilon$ **or** $t \geq T$
- 8: **Return:** $\hat{\theta} = \theta^{(t)}$

In practice, for large n , the gradient $\nabla_{\theta} J(\theta)$ is almost always approximated using a **minibatch** $\mathcal{B}$ of data:

$$\nabla_{\theta} J(\theta) \approx \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \nabla_{\theta} \ell(f(\mathbf{x}_i; \theta), y_i)$$

This variant is called **Stochastic Gradient Descent (SGD)**.

Wheel 5: k-Nearest Neighbors Algorithm

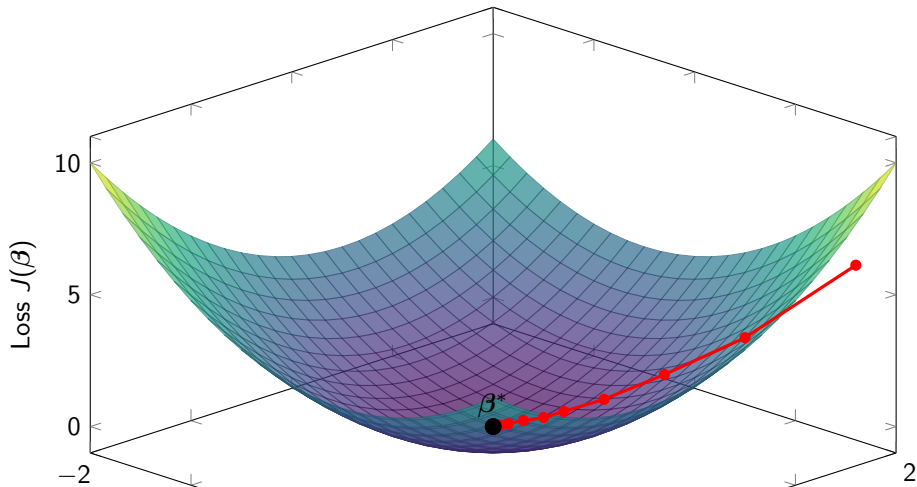
Instance-Based Learning in Action

- 1: **Input:** Training data $\mathcal{D}_n = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$, query point $\mathbf{x}_q$, number of neighbors k , distance metric $d(\cdot, \cdot)$
- 2: **Output:** Predicted label $\hat{y}_q$ for query point
- 3: Initialize empty list: $\text{distances} \leftarrow []$
- 4: **for** $i = 1$ **to** n **do**
 - 5: $\text{distance} \leftarrow d(\mathbf{x}_q, \mathbf{x}_i)$ ▷ Compute distance to each training point
 - 6: $\text{distances.append}((\text{distance}, y_i))$ ▷ Store distance and label
- 7: **end for**
- 8: $\text{sorted_distances} \leftarrow \text{sort}(\text{distances})$ ▷ Sort by ascending distance
- 9: $\text{neighbors} \leftarrow \text{sorted_distances}[1 : k]$ ▷ Extract k nearest neighbors
- 10: **if** regression task **then**
 - 11: $\hat{y}_q \leftarrow \frac{1}{k} \sum_{(\cdot, y) \in \text{neighbors}} y$ ▷ Average for regression
- 12: **else if** classification task **then**
 - 13: $\hat{y}_q \leftarrow \text{mode}(\{y : (\cdot, y) \in \text{neighbors}\})$ ▷ Majority vote for classification
- 14: **end if**
- 15: **Return:** $\hat{y}_q$

Wheel 5: Gradient Descent Optimization

Navigating the Loss Landscape

Gradient Descent: Parameter Space Trajectory



Wheel 6: The Art of Refinement & Stabilization

From Ill-Posed Problems to Well-Regularized Solutions

The Hadamard Revelation

A problem is **well-posed** (au sens d'Hadamard) if:

- 1 **Existence:** A solution exists
- 2 **Uniqueness:** The solution is unique
- 3 **Stability:** The solution depends continuously on the data

Most machine learning problems are **ill-posed**—violating stability, often uniqueness, sometimes existence.

Why ML is Ill-Posed:

- **Finite data, infinite hypotheses**
- **High-dimensional spaces** ($p \gg n$)
- **Non-unique solutions** (e.g., underdetermined systems)
- **Noise amplification** in inverse problems

The Regularization Trinity:

- 1 **Explicit:** Add penalty $\lambda\Omega(f)$ to objective
- 2 **Implicit:** Early stopping, dropout, bagging
- 3 **Data-driven:** Cross-validation, information criteria

$$\hat{f}_{\text{reg}} = \arg \min_{f \in \mathcal{H}} \left[\hat{R}_n(f) + \lambda\Omega(f) \right]$$

Wheel 6: The Regularization Zoo

A Menagerie of Stabilization Techniques

Explicit Regularizers

- L_2 : $\Omega(f) = \|\mathbf{w}\|_2^2$ (Ridge)
- L_1 : $\Omega(f) = \|\mathbf{w}\|_1$ (LASSO)
- Elastic Net:
 $\alpha\|\mathbf{w}\|_1 + (1 - \alpha)\|\mathbf{w}\|_2^2$
- Dantzig: $\|\mathbf{X}^T(\mathbf{y} - \mathbf{X}\mathbf{w})\|_\infty$
- SCAD: Non-convex penalty
- Group LASSO: $\sum_g \|\mathbf{w}_g\|_2$

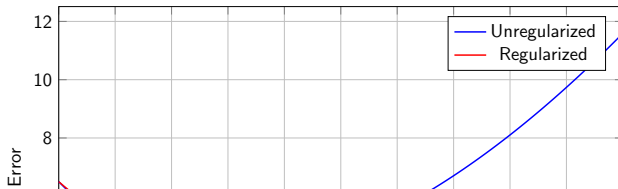
Implicit Regularizers

- Early stopping
- Dropout (neural nets)
- Bagging / Bootstrap
- Noise injection
- Label smoothing
- Data augmentation

Model Selection

- Cross-validation (V-fold, LOOCV)
- Information Criteria (AIC, BIC)
- Minimum Description Length
- Stability selection
- Bayesian model averaging

Regularization's Blessing: The Sweet Spot



Wheel 6: The Ridge-LASSO-ElasticNet Trinity

Convex Regularization in Linear Models

The Regularized Linear Regression Family:

$$\hat{\beta}(\lambda, \alpha) = \arg \min_{\beta} \left\{ \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\beta\|^2 + \lambda \left[\alpha \|\beta\|_1 + \frac{1}{2}(1 - \alpha) \|\beta\|_2^2 \right] \right\}$$

Ridge ($\alpha = 0$)

LASSO ($\alpha = 1$)

Elastic Net ($0 < \alpha < 1$)

$$\hat{\beta}_{\text{ridge}} = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}$$

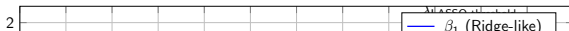
$$\hat{\beta}_{\text{lasso}} = \arg \min_{\beta} \left\{ \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\beta\|^2 + \lambda \|\beta\|_1 \right\}$$

- Shrinks coefficients smoothly
- Never sets coefficients to zero
- Closed-form solution
- Bayesian: Gaussian prior

- Sparse solutions
- Feature selection
- No closed form
- Bayesian: Laplace prior

- Groups correlated variables
- Handles $p > n$ better
- Compromise between Ridge/LASSO
- Bayesian: Normal-Laplace prior

Regularization Paths: How Coefficients Evolve with λ



Wheel 6: The Dantzig Selector

An Honored Guest at Our Regularization Banquet

Theorem (Dantzig Selector (Candes & Tao, 2007))

For linear model $\mathbf{y} = \mathbf{X}\beta^* + \epsilon$ with $\|\epsilon\|_\infty \leq \sigma$, the Dantzig selector solves:

$$\min_{\beta} \|\beta\|_1 \quad \text{subject to} \quad \|\mathbf{X}^T(\mathbf{y} - \mathbf{X}\beta)\|_\infty \leq \lambda\sigma$$

Under Restricted Isometry Property (RIP), with high probability:

$$\|\hat{\beta} - \beta^*\|_2^2 \leq C \cdot s \cdot \sigma^2 \log p$$

where $s = \|\beta^*\|_0$ is sparsity level.

Key Properties:

- Constrains **correlation of residuals** with features
- **Geometric interpretation:** β in ℓ_1 -ball with small correlation constraint

Comparison with LASSO:

	LASSO	Dant
Objective	$\frac{1}{2}\ \mathbf{y} - \mathbf{X}\beta\ ^2$	$\ \beta\ _1$

Wheel 6: The Cross-Validation Symphony

From Stone's LOOCV to Modern V-Fold CV

The Stone Revelation (1974) Leave-One-Out Cross-Validation (LOOCV):

$$CV_{(1)} = \frac{1}{n} \sum_{i=1}^n \left(y_i - \hat{f}^{(-i)}(\mathbf{x}_i) \right)^2$$

where $\hat{f}^{(-i)}$ is trained on all but the i -th observation.

Properties:

- **Almost unbiased** for prediction error
- **High variance** (correlated training sets)
- **Computationally heavy**: n models to fit
- **Exact formula** for linear models:

$$CV_{(1)} = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{y}_i}{1 - h_{ii}} \right)^2$$

V-Fold Cross-Validation Partition data into V folds $\{\mathcal{F}_v\}_{v=1}^V$:

$$CV_{(V)} = \frac{1}{V} \sum_{v=1}^V \frac{1}{|\mathcal{F}_v|} \sum_{i \in \mathcal{F}_v} \left(y_i - \hat{f}^{(-\mathcal{F}_v)}(\mathbf{x}_i) \right)^2$$

Properties:

- **Less expensive**: Fit only V models
- **Bias-variance tradeoff**: $V = 5$ or 10 often optimal
- **Stratification** for classification
- **Monte Carlo CV**: Random splits average

Generalized CV (GCV): Approximation to LOOCV:

$$\frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{y}_i}{1 - h_{ii}} \right)^2$$

Wheel 6: Information Criteria & Model Selection

Beyond Cross-Validation

Akaike Information Criterion (AIC)

$$AIC = 2k - 2 \log \mathcal{L}(\hat{\theta})$$

where k = number of parameters.

Derivation: Approximates KL divergence between true and fitted models.

Properties:

- **Asymptotically efficient:** Minimizes prediction error
- **Not consistent:** Doesn't select true model with probability 1 as $n \rightarrow \infty$
- **Penalty:** $2k$

Bayesian Information Criterion (BIC/SIC)

$$BIC = k \log n - 2 \log \mathcal{L}(\hat{\theta})$$

Other Notable Criteria:

Mallows' C_p :

$$C_p = \frac{RSS}{s^2} + 2p - n$$

For linear models, where s^2 is MSE of full model.

Adjusted R^2 :

$$R_{adj}^2 = 1 - \frac{(1 - R^2)(n - 1)}{n - p - 1}$$

Penalizes adding useless predictors.

Minimum Description Length (MDL):

$$MDL = -\log \mathcal{L}(\hat{\theta}) + \frac{k}{2} \log n + \log \int \sqrt{\det I(\theta)} d\theta$$

Wheel 6: Aggregation & Ensemble Methods

The Wisdom of Committees

The Ensemble Principle

Multiple weak learners can be combined to create a strong learner, often outperforming any single constituent.

Bagging (Bootstrap Aggregating)

- Train on bootstrap samples
- Average predictions (regression)
- Majority vote (classification)
- Reduces variance
- Example: Random Forests

$$\hat{f}_{\text{bag}}(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^B \hat{f}^{*b}(\mathbf{x})$$

Boosting

Sequentially fit residuals
Weight misclassified examples more
Creates strong learner from weak ones
Example: AdaBoost, Gradient Boosting

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \nu \cdot h_m(\mathbf{x})$$

where h_m fits residuals.

Stacking

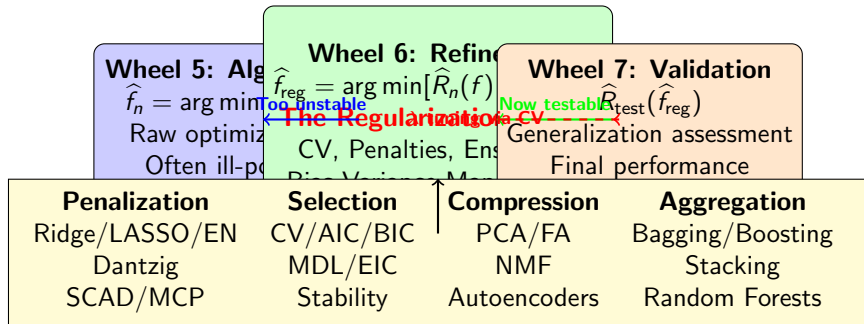
- Learn optimal combination of base learners
- Meta-learner on base predictions
- Can use CV to prevent overfitting
- Often best performance

$$\hat{f}_{\text{stack}}(\mathbf{x}) = g\left(\hat{f}_1(\mathbf{x}), \dots, \hat{f}_M(\mathbf{x})\right)$$

where g is learned.

Wheel 6: The Grand Synthesis

Regularization as the Bridge Between Wheels 5 and 7



Why Wheel 6 is Essential:

- Tames ill-posedness (Hadamard)
- Manages bias-variance tradeoff
- Enables interpretability via simplification
- Improves generalization to Wheel 7

The Regularization Mindset:

"Better to be approximately right than precisely wrong."
— John Tukey (paraphrased)

Regularization embraces approximation as a strategic compromise, trading mathematical

Wheel 6: Refinement & Stabilization

Taming Overfitting: The Art of Regularization

“The true method of knowledge is experiment.” – Jacques Hadamard

Regularization is any technique that reduces overfitting (variance) at the cost of a small increase in bias.

- L_2 (**Ridge/Tikhonov**): Penalizes large weights: $\Omega(\mathbf{w}) = \|\mathbf{w}\|_2^2$. Shrinks coefficients smoothly.
- L_1 (**Lasso**): Penalizes absolute value: $\Omega(\mathbf{w}) = \|\mathbf{w}\|_1$. Promotes sparsity (feature selection).
- **Dropout**: Randomly “drops” neurons during DNN training. Prevents co-adaptation. Like training an ensemble of thinned networks.
- **Early Stopping**: A free and highly effective regularizer! Stop training when validation error stops decreasing. Prevents the model from over-optimizing on the training noise.

Wheel 6: Refinement & Stabilization

Regularization and Validation

To prevent overfitting, we minimize the **Regularized Empirical Risk**:

$$\hat{f} = \arg \min_{f \in \mathcal{H}} \left[\hat{R}_n(f) + \lambda \Omega(f) \right]$$

where $\Omega(f)$ penalizes complexity (e.g., $\Omega(f) = \|\mathbf{w}\|_2^2$ for Ridge, $\|\mathbf{w}\|_1$ for Lasso) and λ is a **hyperparameter**.

How to choose λ ? Use **V-Fold Cross-Validation (CV)**—the Swiss Army knife of data mining.

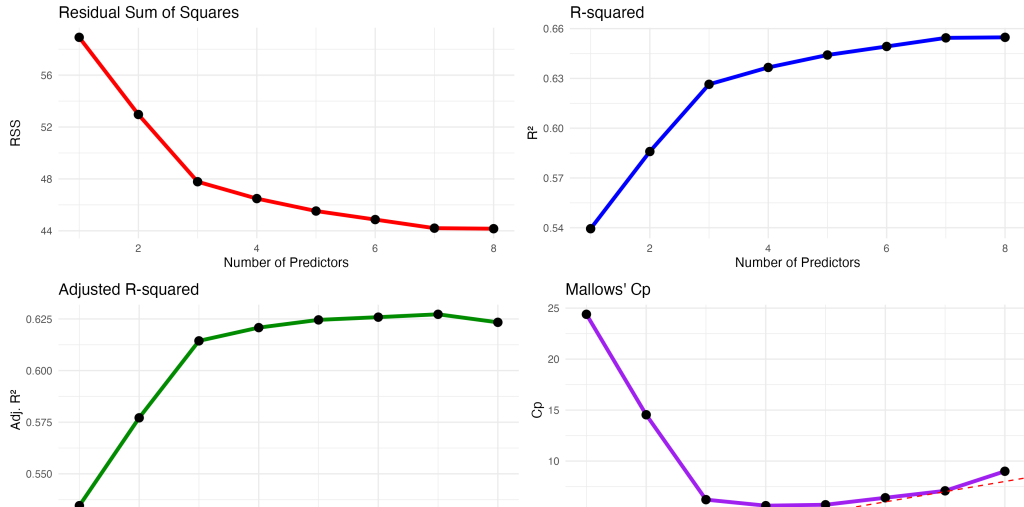
$$\hat{R}_n^{\text{CV}}(\lambda) = \frac{1}{V} \sum_{v=1}^V \hat{R}_n^{(v)}(f^{-\mathcal{F}_v; \lambda})$$

- Split data into V folds $\{\mathcal{F}_v\}_{v=1}^V$.
- For each v : Train model $f^{-\mathcal{F}_v; \lambda}$ on all data *except* fold $\mathcal{F}_v$.
- Compute the risk $\hat{R}_n^{(v)}$ on the held-out fold $\mathcal{F}_v$.
- The CV score $\hat{R}_n^{\text{CV}}(\lambda)$ estimates the generalization error for λ .
- Choose $\lambda^* = \arg \min_{\lambda} \hat{R}_n^{\text{CV}}(\lambda)$.

Wheel 6: Model Selection Criteria

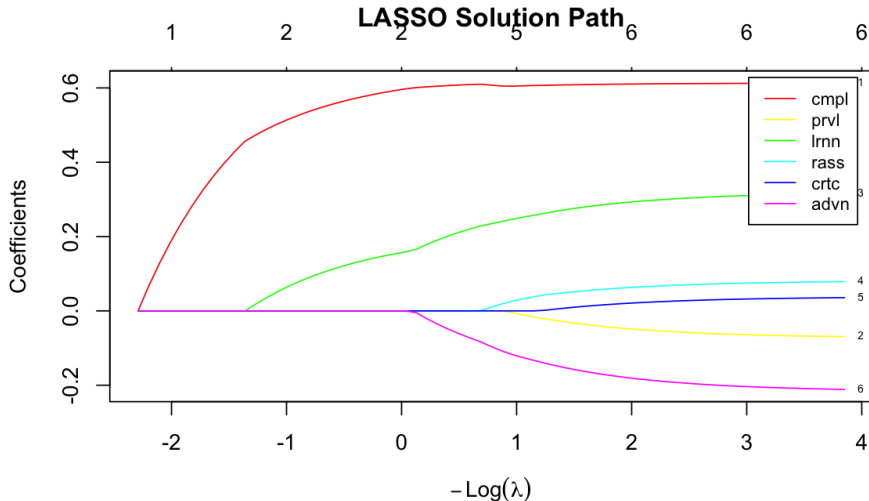
Balancing Fit and Complexity

Model Selection Criteria Comparison



Wheel 6: LASSO Regularization

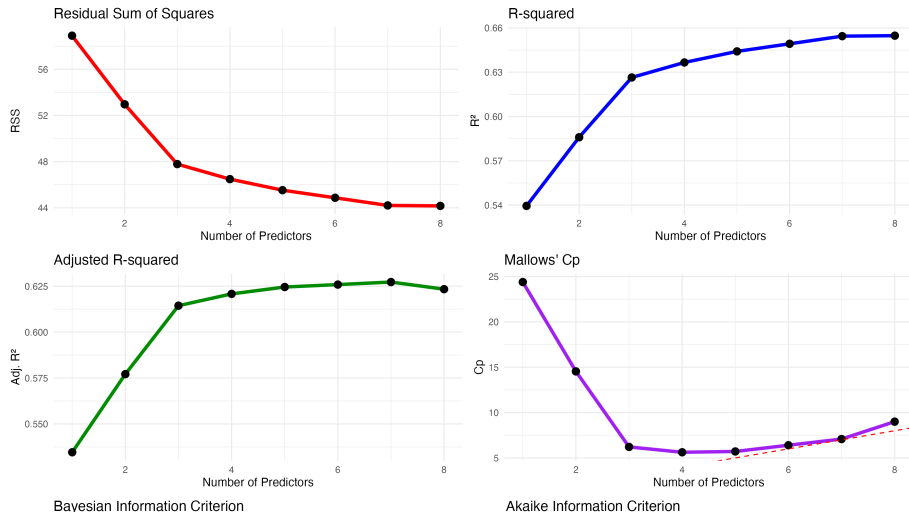
Sparse Solutions via L1 Penalization



Wheel 6: Model Selection Criteria

Balancing Fit and Complexity

Model Selection Criteria Comparison

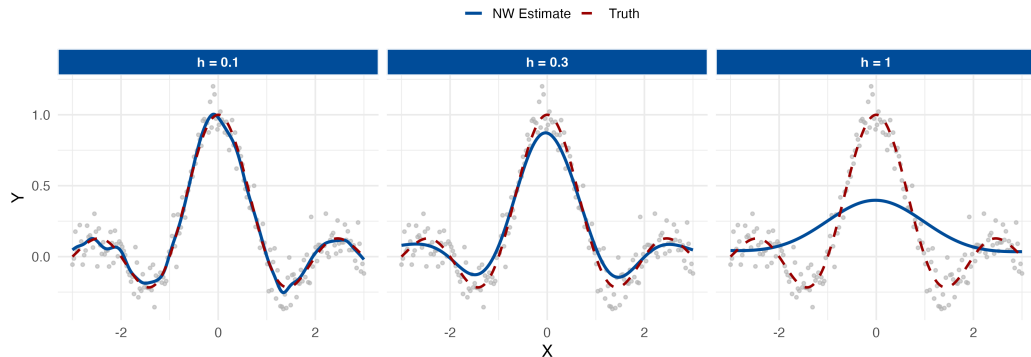


Wheel 6: Bandwidth Selection in Kernel Regression

The Bias-Variance Tradeoff in Practice

Nadaraya-Watson Regression: Bandwidth Selection

Effect of different bandwidth values on estimation quality



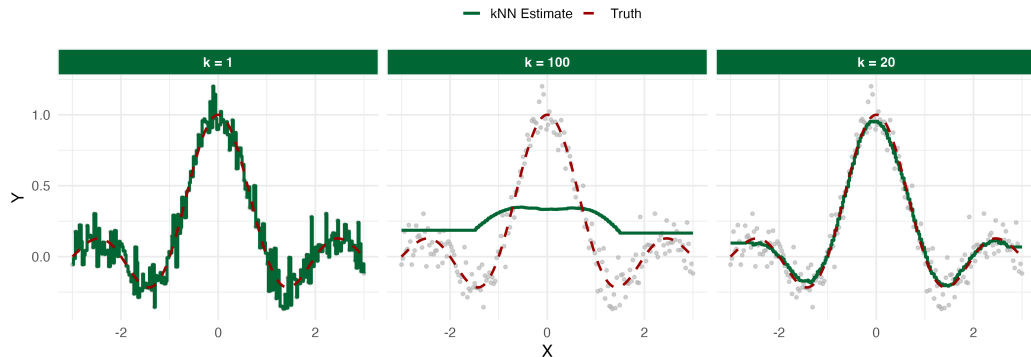
- **Small bandwidth ($h=0.1$):** High variance, follows noise
- **Medium bandwidth ($h=0.3$):** Good balance, captures signal
- **Large bandwidth ($h=1$):** High bias, oversmoothed

Wheel 6: k-NN Neighborhood Size Selection

Model Complexity through Neighborhood Choice

k-Nearest Neighbors Regression

Effect of different k values on estimation quality



- **Small k (k=1):** High variance, perfect interpolation, overfits
- **Medium k (k=20):** Balanced, good generalization
- **Large k (k=100):** High bias, oversmoothed, underfits

Wheel 7: Extrinsic Validation & Predictive Comparisons

The Empirical Court of Performance Evaluation

The No-Free-Lunch Reality

“There is no universally superior learning algorithm.”

The No-Free-Lunch theorems (Wolpert, 1996) establish that averaging over *all* possible problems, every algorithm performs equally.

Thus: We need **empirical benchmarking** on *specific* problems to compare algorithms.

The Validation Imperative:

- **Intrinsic:** Train-test split on *same* algorithm
- **Extrinsic:** Compare *different* algorithms
- **Relative:** How good is *relative* to alternatives?
- **Absolute:** How good is *relative* to task requirements?

Performance Metrics Zoo:

- **Regression:** MSE, MAE, R^2 , MAPE
- **Classification:** Accuracy, F1, AUC-ROC, Precision, Recall
- **Ranking:** NDCG, MRR, Precision@k
- **Probability:** Log-loss, Brier score, Calibration

$\mathbb{1}_{\{\text{Algorithm A beats B}\}} = f(\text{Performance}_A, \text{Performance}_B)$ • **Business:** Profit, Retention, Conversion

Wheel 7: The Stochastic Holdout Procedure

Monte Carlo Estimation of Comparative Performance

- 1: **Input:** Full dataset $\mathcal{D}_n$, number of splits S , test fraction α , algorithms $\{\mathcal{A}_m\}_{m=1}^M$
- 2: **Initialize:** Performance matrix $\mathbf{P} \in \mathbb{R}^{S \times M}$, $\mathbf{P} \leftarrow \mathbf{0}$
- 3: **for** $s = 1$ **to** S **do**
- 4: Random split: $\mathcal{D}_n \rightarrow (\mathcal{D}_{\text{tr}}^{(s)}, \mathcal{D}_{\text{te}}^{(s)})$ with $|\mathcal{D}_{\text{te}}^{(s)}| = \lfloor \alpha n \rfloor$
- 5: **for** $m = 1$ **to** M **do**
- 6: Train: $\hat{f}_m^{(s)} \leftarrow \mathcal{A}_m(\mathcal{D}_{\text{tr}}^{(s)})$
- 7: Test: $p_{sm} \leftarrow \text{Metric}(\hat{f}_m^{(s)}, \mathcal{D}_{\text{te}}^{(s)})$
- 8: Store: $\mathbf{P}[s, m] \leftarrow p_{sm}$
- 9: **end for**
- 10: **end for**
- 11: **Output:** Performance matrix $\mathbf{P}$, summary statistics, visualizations

Mathematical Formulation: For algorithm m
on split s :

$$p_{sm} = \frac{1}{|\mathcal{D}_{\text{te}}^{(s)}|} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}_{\text{te}}^{(s)}} \phi\left(y_i, \hat{f}_m^{(s)}(\mathbf{x}_i)\right)$$

Key Properties:

- **Variance reduction:** S estimates average out split randomness
- **Bias:** Slight if α small (less training data)
- **Computational cost:** $S \times M$ training

Wheel 7: Performance Matrix Analysis

From Raw Scores to Statistical Inference

The Performance Matrix $\mathbf{P} \in \mathbb{R}^{S \times M}$

$$\mathbf{P} = \begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1M} \\ p_{21} & p_{22} & \cdots & p_{2M} \\ \vdots & \vdots & \ddots & \vdots \\ p_{S1} & p_{S2} & \cdots & p_{SM} \end{pmatrix}$$

Rows: Different data splits (S), Columns: Different algorithms (M)

Column Summaries (per algorithm):

Mean: $\bar{p}_{\cdot m} = \frac{1}{S} \sum_{s=1}^S p_{sm}$

Std: $s_m = \sqrt{\frac{1}{S-1} \sum_{s=1}^S (p_{sm} - \bar{p}_{\cdot m})^2}$

SE: $SE_m = s_m / \sqrt{S}$

Pairwise Comparisons:

$$\Delta_{m,m'} = \frac{1}{S} \sum_{s=1}^S (p_{sm} - p_{sm'})$$

$$SE(\Delta_{m,m'}) = \sqrt{\frac{1}{S(S-1)} \sum_{s=1}^S [(p_{sm} - p_{sm'}) - \Delta_{m,m'}]^2}$$

Wheel 7: Statistical Inference for Algorithm Comparison

Beyond Descriptive Statistics

The Fundamental Question

When algorithm A outperforms algorithm B on average, is the difference **statistically significant** or just random variation?

Paired Statistical Tests: Since p_{sm} and $p_{sm'}$ come from same split s , they are *paired*.

Paired t-test:

$$t = \frac{\bar{d}}{s_d / \sqrt{S}} \sim t_{S-1}$$

where $d_s = p_{sm} - p_{sm'}$, $\bar{d} = \frac{1}{S} \sum d_s$,
 $s_d^2 = \frac{1}{S-1} \sum (d_s - \bar{d})^2$.

Wilcoxon Signed-Rank Test: Non-parametric alternative, tests median difference.

Multiple Comparisons: When comparing M algorithms, we have $\binom{M}{2}$ pairwise tests.

Corrections needed:

- Bonferroni: $\alpha_{\text{adj}} = \alpha / \binom{M}{2}$
- Holm-Bonferroni: Step-down procedure
- Benjamini-Hochberg: Control False Discovery Rate

ANOVA for Algorithms:

$$F = \frac{\text{Between-algorithm variance}}{\text{Within-algorithm variance}} \sim F_{M-1, S(M-1)}$$

Wheel 7: Beyond Stochastic Holdout

Theoretically Grounded Alternatives

Cross-Validation Variants:

- **Nested CV:** Outer loop for evaluation, inner for tuning
- **Stratified CV:** Maintain class proportions
- **Group/Cluster CV:** Respect data dependencies
- **Time Series CV:** Forward chaining for temporal data

The Bootstrap: Draw B bootstrap samples $\mathcal{D}_n^{*b}$:

$$\widehat{\text{Err}}_{\text{boot}} = \frac{1}{B} \sum_{b=1}^B \frac{1}{n} \sum_{i=1}^n \phi \left(y_i, \hat{f}^{*b}(\mathbf{x}_i) \right)$$

where $\hat{f}^{*b}$ trained on $\mathcal{D}_n^{*b}$.

Bias-Corrected (".632") Bootstrap:

Theoretical Bounds:

- **VC Bounds:** $R(f) \leq \hat{R}_n(f) + O \left(\sqrt{\frac{d_{\text{VC}}}{n}} \right)$

- **Rademacher Complexity:**

$$R(f) \leq \hat{R}_n(f) + 2\mathfrak{R}_n(\mathcal{H}) + O \left(\sqrt{\frac{\log(1/\delta)}{n}} \right)$$

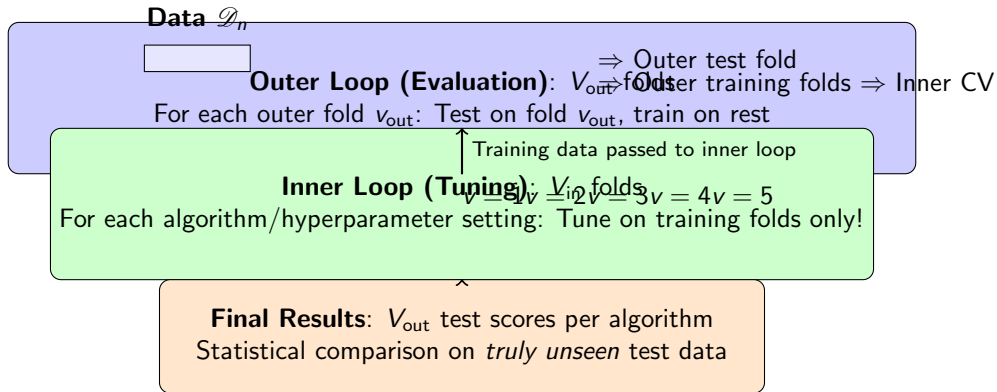
- **PAC-Bayes:** Combines prior knowledge with data
- **Stability Bounds:** Algorithms stable to data perturbations generalize

Information Criteria:

$$\text{AIC} = 2k - 2 \log \mathcal{L}, \quad \text{BIC} = k \log n - 2 \log \mathcal{L}$$

Wheel 7: The Nested Cross-Validation Protocol

The Gold Standard for Fair Algorithm Comparison



Why Nested?

- **Avoids optimistic bias:** Hyperparameter tuning uses only training data

Implementation:

```
for  $v_{out} = 1$  to  $V_{out}$  do  
     $\mathcal{D}_{train}, \mathcal{D}_{test} \leftarrow \text{Split}(\mathcal{D}_n, v_{out})$ 
```

Wheel 7: The Validation Decision Matrix

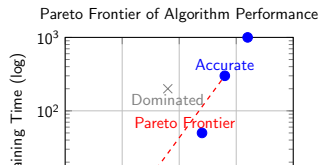
From Empirical Evidence to Practical Decisions

Criterion	Algorithm A	Algorithm B	Algorithm C	Decision Weight
Mean Test Accuracy	0.85 ± 0.02	0.87 ± 0.03	0.89 ± 0.02	40%
Training Time (s)	10.2	45.7	312.5	20%
Inference Time (ms)	2.1	8.7	15.3	15%
Interpretability	High	Medium	Low	15%
Memory Usage (MB)	5.2	12.7	89.3	10%
Weighted Score	82.4	76.8	71.2	100%

Multi-Criteria Decision Making: Performance $\neq$ just accuracy!

- **Statistical:** Accuracy, precision, recall, calibration
- **Computational:** Training/inference time, memory
- **Practical:** Interpretability, maintainability
- **Ethical:** Fairness, bias, privacy

The Pareto Frontier: An algorithm is **Pareto-optimal** if no other algorithm is better on *all* criteria.



Wheel 7: Extrinsic Validation

The Stochastic Holdout

The simplest yet most crucial validation technique is the **Holdout Method**.

- **Training Set ($\mathcal{D}_{\text{train}}$):** Used to solve the arg min for the model parameters θ .
- **Validation Set ($\mathcal{D}_{\text{val}}$):** Used to tune hyperparameters (like λ) and model selection. (*Often replaced by CV.*)
- **Test Set ($\mathcal{D}_{\text{test}}$):** Used **only once** to compute a final, unbiased estimate of the generalization error $R(f)$.

$$\hat{R}_{\text{test}}(f) = \frac{1}{|\mathcal{D}_{\text{test}}|} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}_{\text{test}}} \ell(f(\mathbf{x}_i), y_i)$$

- The test set must remain **locked away** until the very end of all development. It is the final exam.

Wheel 7: Extrinsic Validation

The Monte Carlo Holdout

"Prediction is very difficult, especially if it's about the future." – Niels Bohr

- 1: **Given:** Full dataset $\mathcal{D}_n$, number of replicates S , test set fraction α
- 2: Initialize vector of test errors: $\mathbf{e}_{\text{test}} = []$ ▷ An empty list
- 3: **for** $s = 1$ **to** S **do**
- 4: $\mathcal{D}_{\text{train}}^{(s)}, \mathcal{D}_{\text{test}}^{(s)} \leftarrow \text{Split}(\mathcal{D}_n, \alpha)$ ▷ Randomly split data
- 5: $\hat{f}^{(s)} \leftarrow \arg \min_{f \in \mathcal{H}} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}_{\text{train}}^{(s)}} \ell(f(\mathbf{x}_i), y_i)$ ▷ Train on training fold
- 6: $e_{\text{test}}^{(s)} \leftarrow \frac{1}{|\mathcal{D}_{\text{test}}^{(s)}|} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}_{\text{test}}^{(s)}} \ell(\hat{f}^{(s)}(\mathbf{x}_i), y_i)$ ▷ Compute test error
- 7: $\mathbf{e}_{\text{test}}.append(e_{\text{test}}^{(s)})$
- 8: **end for**
- 9: **Return:** $\hat{R}_{\text{test}} = \frac{1}{S} \sum_{s=1}^S e_{\text{test}}^{(s)}, \quad \widehat{SE}(\hat{R}_{\text{test}}) = \sqrt{\frac{1}{S(S-1)} \sum_{s=1}^S (e_{\text{test}}^{(s)} - \hat{R}_{\text{test}})^2}$

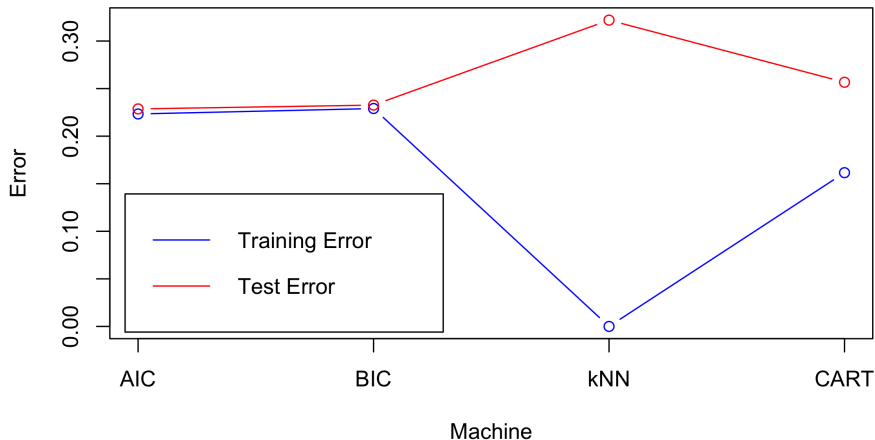
The test error for a single split s can be written compactly using the indicator function:

$$e_{\text{test}}^{(s)} = \frac{\sum_{i=1}^n \ell(\hat{f}^{(s)}(\mathbf{x}_i), y_i) \cdot \mathbb{1}_{\{(\mathbf{x}_i, y_i) \in \mathcal{D}_{\text{test}}^{(s)}\}}}{\sum_{i=1}^n \mathbb{1}_{\{(\mathbf{x}_i, y_i) \in \mathcal{D}_{\text{test}}^{(s)}\}}}$$

Wheel 7: Training vs Test Error

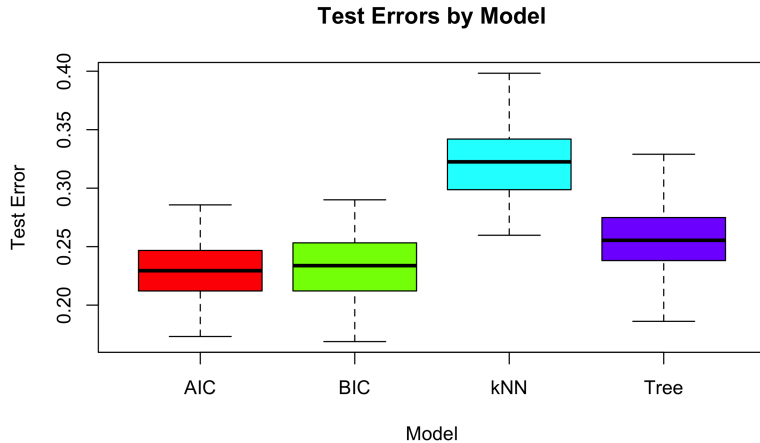
The Reality Check for Generalization

Average Training and Test Errors



Wheel 7: Comparative Model Performance

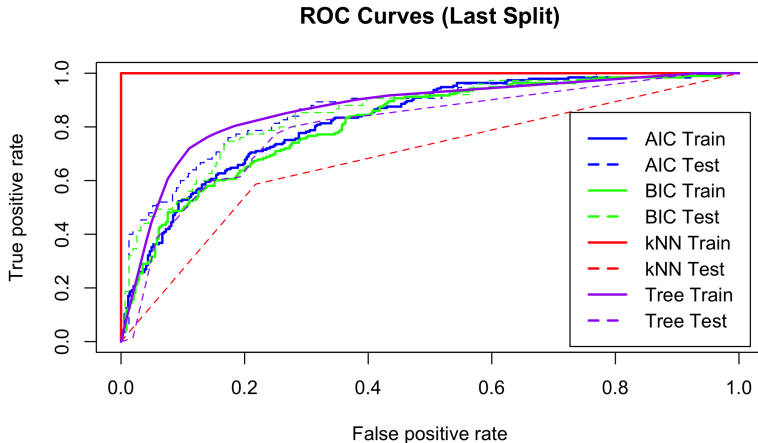
Test Errors Across Different Approaches



• Boxplots show distribution of test errors across multiple runs

Wheel 7: ROC Curve Analysis

Evaluating Classification Performance



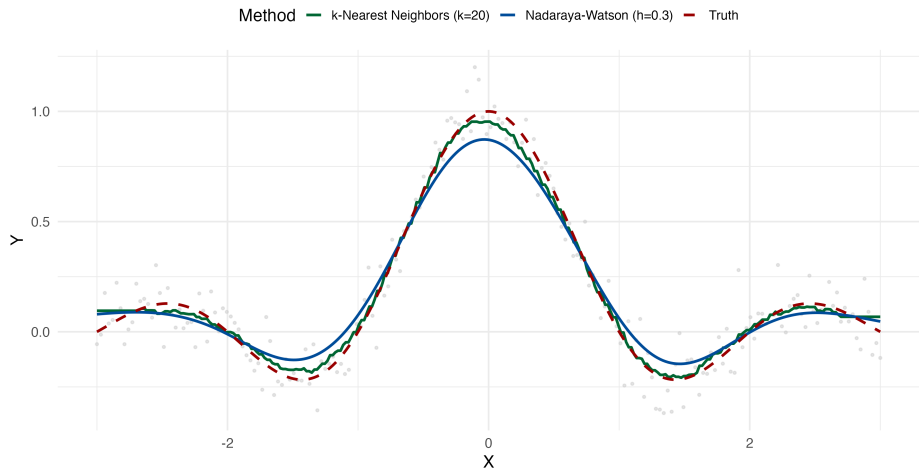
ROC curves visualize tradeoff between true positive and false positive rates

Wheel 7: Nonparametric Regression Methods

k-NN vs Nadaraya-Watson Kernel Regression

Nonparametric Regression: kNN vs Nadaraya-Watson

Comparison of optimal kernel and nearest neighbors estimates



Wheel 8: Theoretical Foundations & Statistical Inference

From Empirical Observations to Provable Guarantees

The Sacred Quest for Certainty

“In God we trust, all others must bring data.”

— W. Edwards Deming

But how much data? And what guarantees can we provide?

The Inference Trinity:

- ① **Point Estimation:** $\hat{\theta}_n$ as guess of θ
- ② **Confidence Sets:** C_n with
 $\mathbb{P}(\theta \in C_n) \geq 1 - \alpha$
- ③ **Hypothesis Testing:** $H_0 : \theta \in \Theta_0$ vs
 $H_1 : \theta \in \Theta_1$

For Learning Machines:

$\hat{f}_n \rightarrow f^*$ with guarantees on $R(\hat{f}_n) - R(f^*)$

The Concentration Philosophy:

“Sample means concentrate around population means.

The universe hates outliers as much as statisticians do.”

$$\mathbb{P}\left(|\hat{\theta}_n - \theta| \geq \epsilon\right) \xrightarrow{n \rightarrow \infty} 0$$

The rate of convergence reveals the **price of certainty**.

Wheel 8a: The Concentration Pantheon

From Humble Beginnings to Exponential Magic

The Classics Markov ($X \geq 0$):

$$\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}[X]}{t}$$

Chebyshev: Uses variance σ^2 for $1/t^2$ decay:

$$\mathbb{P}(|X - \mu| \geq t) \leq \frac{\sigma^2}{t^2}$$

Chernoff (Bernoulli): Specific to binary sums:

$$\mathbb{P}(\bar{X}_n - p \geq \epsilon) \leq e^{-nD(p+\epsilon||p)}$$

Modern Learning Tools Hoeffding (Bounded): For

$X_i \in [a, b]$, provides exp. decay:

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \epsilon) \leq 2e^{-\frac{2n\epsilon^2}{(b-a)^2}}$$

Bernstein (Variance-Aware): Tighter than Hoeffding if $\sigma^2 \ll 1$:

$$\mathbb{P}(\dots) \leq 2 \exp\left(-\frac{n\epsilon^2}{2\sigma^2 + \frac{2}{3}M\epsilon}\right)$$

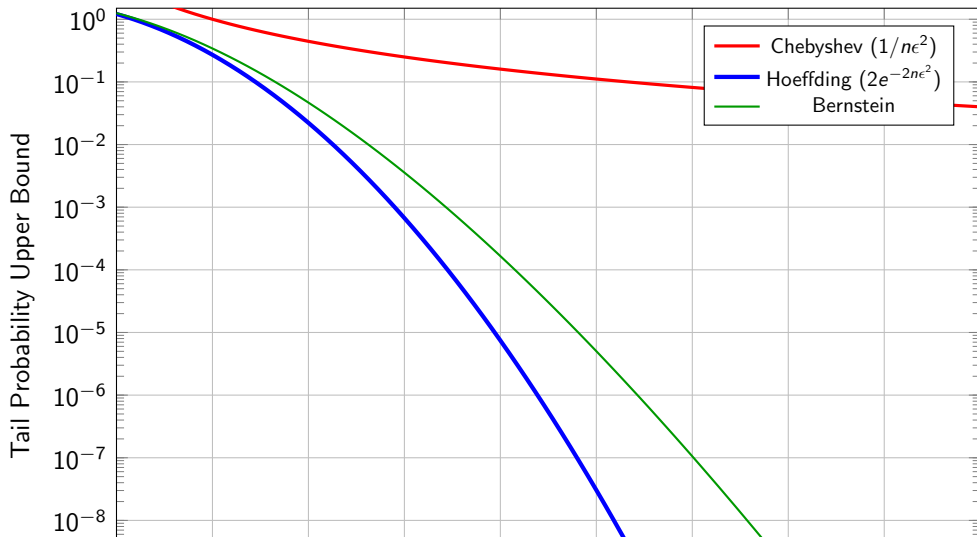
McDiarmid: For any f with bounded differences c_i :

$$\mathbb{P}(|f - \mathbb{E}f| \geq \epsilon) \leq 2e^{-\frac{2\epsilon^2}{\sum c_i^2}}$$

Key Takeaway

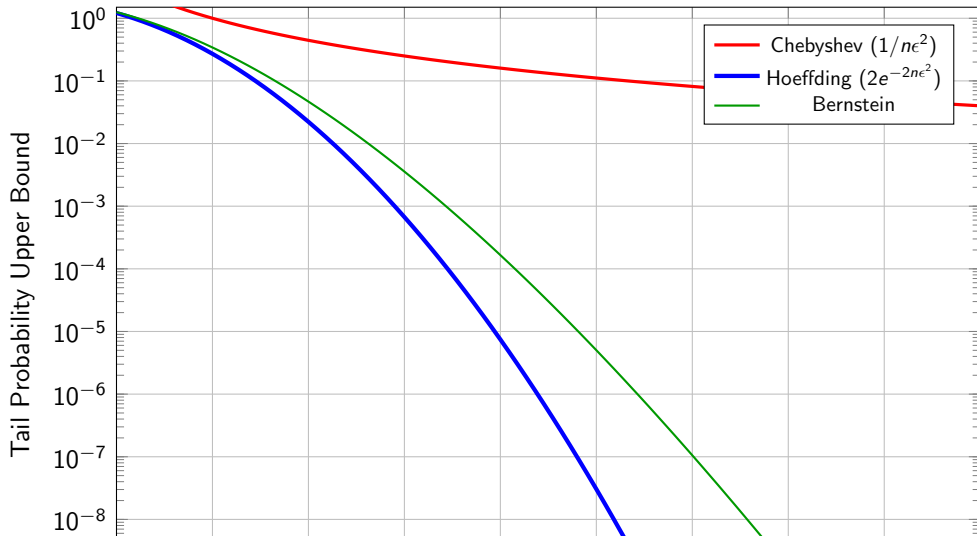
Wheel 8b: Visualizing Concentration

Polynomial vs. Exponential Decay ($n = 100$)



Wheel 8b: Visualizing Concentration

Polynomial vs. Exponential Decay ($n = 100$)



Wheel 8a: Rademacher Complexity

A Modern Measure of Function Class Richness

Definition (Empirical Rademacher Complexity)

For function class $\mathcal{F}$ and data $S = \{x_1, \dots, x_n\}$:

$$\hat{\mathfrak{R}}_S(\mathcal{F}) = \mathbb{E}_{\sigma} \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \sigma_i f(x_i) \right]$$

where $\sigma_i \stackrel{\text{i.i.d.}}{\sim} \text{Uniform}\{\pm 1\}$ are Rademacher variables.

The Fundamental Generalization Bound: With probability at least $1 - \delta$, for all $f \in \mathcal{F}$:

$$R(f) \leq \hat{R}_n(f) + 2\hat{\mathfrak{R}}_S(\mathcal{F}) + 3\sqrt{\frac{\log(2/\delta)}{2n}}$$

Why it matters

Unlike VC-dimension, Rademacher complexity is **data-dependent**. It measures the complexity of $\mathcal{F}$

Wheel 8b: Complexity Intuition

Correlating with Noise

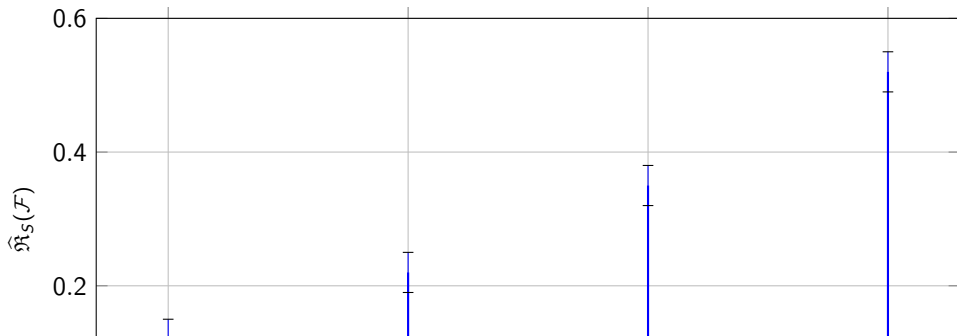
Interpretation:

- Ability to **correlate with random noise**
- **Large** $\mathfrak{R}_n$: Overfits easily
- **Small** $\mathfrak{R}_n$: Stronger regularization

Key Properties:

- Linear class: $\mathfrak{R}_n \leq \frac{R\|\mathbf{w}\|}{\sqrt{n}}$
- Subadditive: $\mathfrak{R}_n(\mathcal{F} + \mathcal{G}) \leq \mathfrak{R}_n(\mathcal{F}) + \mathfrak{R}_n(\mathcal{G})$

Empirical Complexity vs. Model Class



Wheel 8a: Uniform Convergence

From Fixed Functions to Learned Models

The Fundamental Challenge

Hoeffding bounds **one fixed** f :

$$\mathbb{P}(|\hat{R}_n(f) - R(f)| \geq \epsilon) \leq 2e^{-2n\epsilon^2}$$

But the ERM optimizer $\hat{f}_n$ depends on data! We need the bound to hold for **all** $f \in \mathcal{H}$ simultaneously.

Theorem (Vapnik-Chervonenkis Inequality)

For a class $\mathcal{H}$ with VC dimension d , with probability at least $1 - \delta$:

$$\sup_{f \in \mathcal{H}} |R(f) - \hat{R}_n(f)| \leq \sqrt{\frac{2}{n} \left(d \ln \frac{2en}{d} + \ln \frac{2}{\delta} \right)}$$

- **Generalization Gap:** The difference between training and test error.
- **Uniformity:** The bound holds for the worst-case function in the class.

Wheel 8b: VC Dimension & Shattering

Measuring the Combinatorial Richness of $\mathcal{H}$

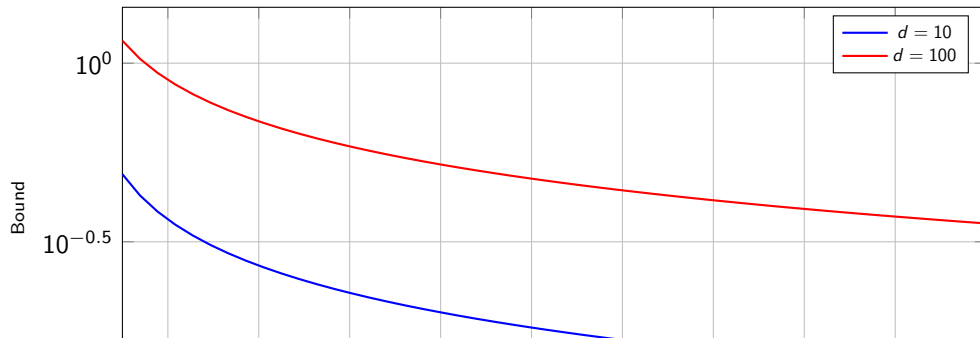
The Shattering Concept $d_{VC}(\mathcal{H})$ is the largest m such that $\mathcal{H}$ can realize all 2^m labelings on m points.

Sauer-Shelah Lemma: If $d_{VC} = d$, the growth function $\Pi_{\mathcal{H}}(m)$ is polynomial, not exponential:

$$\Pi_{\mathcal{H}}(m) \leq \left(\frac{em}{d}\right)^d$$

Examples of d_{VC} :

- **Linear** ($x \in \mathbb{R}^p$): $p + 1$
- **Intervals**: 2
- **1NN Classifier**: ∞
- **Convex Polygons**: ∞



Wheel 8a: Generalization Bounds in Action

Decomposing the Error of the ERM Estimator

Theorem (Generalization Bound via Rademacher)

With probability at least $1 - \delta$, for all $f \in \mathcal{H}$:

$$R(f) \leq \hat{R}_n(f) + 2\mathfrak{R}_n(\ell \circ \mathcal{H}) + \sqrt{\frac{\ln(2/\delta)}{2n}}$$

Excess Risk Decomposition: The total error of our learned model $\hat{f}_n$ splits into two parts:

$$\underbrace{R(\hat{f}_n) - R(f^*)}_{\text{Total Excess Risk}} \leq \underbrace{2\mathfrak{R}_n(\ell \circ \mathcal{H}) + \mathcal{O}\left(\sqrt{\frac{1}{n}}\right)}_{\text{Estimation Error (Variance)}} + \underbrace{[R(f_{\mathcal{H}}^*) - R(f^*)]}_{\text{Approximation Error (Bias)}}$$

- **Estimation Error:** Decreases as $n \rightarrow \infty$ (statistical price).
- **Approximation Error:** Depends on how "rich" $\mathcal{H}$ is (modeling price).

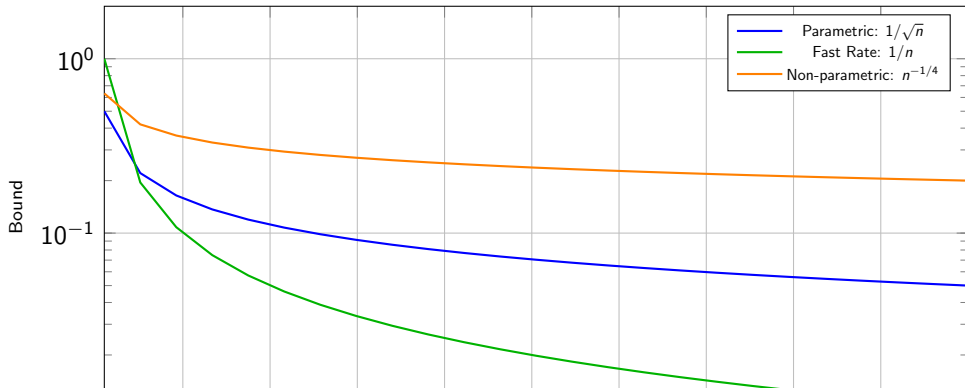
Wheel 8b: Complexity of Model Classes

Learning Rates: How fast do we converge?

Linear Classifiers: $\mathfrak{R}_n(\mathcal{H}) \leq \frac{B \cdot \max \|x_i\|}{\sqrt{n}}$ Rate:
 $\mathcal{O}(1/\sqrt{n})$

Kernel Methods (RKHS): $\mathfrak{R}_n(\mathcal{H}) \leq \frac{\sqrt{\text{Tr}(K)}}{n}$ Rate:
 $\mathcal{O}(1/n)$ (Fast!)

Neural Networks: For L layers and width d :
 $\mathfrak{R}_n \leq \frac{C(B,R)^L \sqrt{d \ln n}}{\sqrt{n}}$ Note: Highly dependent on weight norm B .



Wheel 8: Statistical Inference for Learned Models

Confidence, Testing, and Uncertainty Quantification

Confidence Intervals for Predictions For linear regression $\hat{y} = \mathbf{x}^\top \hat{\beta}$:

$$\hat{y} \pm t_{1-\alpha/2, n-p} \cdot \hat{\sigma} \sqrt{\mathbf{x}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}}$$

- **For mean response:** As above
- **For individual response:** Add $\hat{\sigma}^2$ inside sqrt

For General f via Bootstrap:

- 1 Draw bootstrap samples $\mathcal{D}_n^{*1}, \dots, \mathcal{D}_n^{*B}$
- 2 Compute $\hat{f}^{*1}, \dots, \hat{f}^{*B}$
- 3 Prediction interval: $\alpha/2$ to $1 - \alpha/2$ quantiles of $\{\hat{f}^{*b}(\mathbf{x})\}$

Hypothesis Testing in ML:

- **Feature importance:** $H_0 : \beta_j = 0$
- **Model comparison:** $H_0 : R(f_1) = R(f_2)$
- **Fairness testing:** H_0 : Equal performance across groups
- **Drift detection:** $H_0 : p_{\text{old}}(\mathbf{x}) = p_{\text{new}}(\mathbf{x})$

Likelihood Ratio Tests:

$$\Lambda = \frac{\max_{\theta \in \Theta_0} \mathcal{L}(\theta)}{\max_{\theta \in \Theta} \mathcal{L}(\theta)}$$

Under H_0 : $-2 \log \Lambda \xrightarrow{d} \chi^2_{\dim(\Theta) - \dim(\Theta_0)}$

Wheel 8: The Wisdom of Theoretical Learning

What Theory Teaches Us About Practice

Theoretical Insights:

① The Bias-Variance-Cost Trilemma:

- Simple models: High bias, low variance, low cost
- Complex models: Low bias, high variance, high cost
- Optimal model: Balances all three

② The Sample Complexity Principle:

$$n \geq \frac{C \cdot \text{Complexity}(\mathcal{H})}{\epsilon^2}$$

More complex models need exponentially more data.

③ The Curse of Dimensionality: In $\mathbb{R}^p$, volume grows as r^p — data becomes sparse.

Practical Guidelines from Theory:

- **Start simple:** Linear models have $d_{VC} = p + 1$
- **Regularize:** Controls effective complexity
- **Cross-validate:** Empirical estimate of $\mathfrak{R}_n$
- **Monitor gaps:** $\hat{R}_n - R_{\text{test}}$ indicates overfitting
- **Consider stability:** Stable algorithms generalize

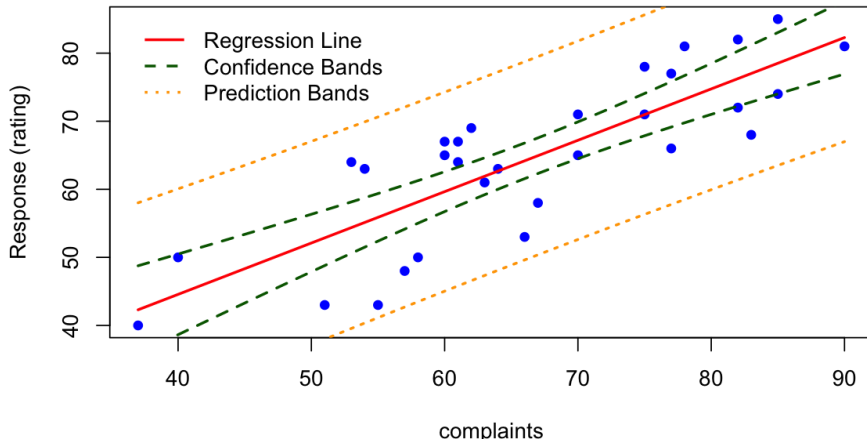
The Fundamental Limits:

- ① **No Free Lunch:** No universally best algorithm
- ② **Computational-Statistical Tradeoffs:** Faster algorithms may need more data
- ③ **Irreducible Error:** $R(f^*) = \sigma^2$ is the floor

Wheel 8: Statistical Inference in Regression

Uncertainty Quantification

Confidence and Prediction Bands for complaints



Wheel 8: Theoretical Guarantees

Generalization Bounds: From Simple to Profound

- **Simple Useful (Hoeffding's Inequality):** For a fixed function f and bounded loss $0 \leq \ell \leq 1$:

$$\mathbb{P}\left(|R(f) - \hat{R}_n(f)| \geq \epsilon\right) \leq 2 \exp(-2n\epsilon^2)$$

Holds with high probability for a **single** f , given enough data (n large).

- **Historic Profound (VC Bound):** For **any** function f in a class $\mathcal{H}$ with finite VC dimension d_{VC} , with probability at least $1 - \delta$:

$$R(f) \leq \hat{R}_n(f) + \sqrt{\frac{2}{n} \left(d_{VC} \ln \left(\frac{2en}{d_{VC}} \right) + \ln \left(\frac{2}{\delta} \right) \right)}$$

This **uniform convergence** bound justifies the ERM principle for the entire class $\mathcal{H}$. It is the cornerstone of Statistical Learning Theory.

The Sacred Turning of Wheel 8

Where Mathematics Becomes Certainty, Where Data Becomes Wisdom

THEOREMA MAGNUM

Theorem (The Statistical Learning Trinity)

Let $\mathcal{H}$ be a hypothesis space with finite complexity. Then with probability at least $1 - \delta$:

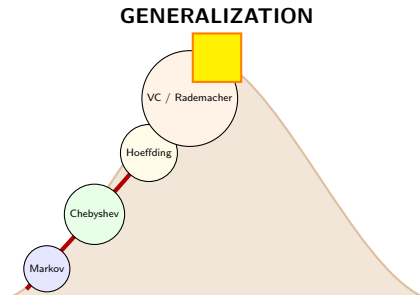
$$\underbrace{R(\hat{f}_n)}_{\text{Risk}} \leq \underbrace{\hat{R}_n(\hat{f}_n)}_{\text{Empirical}} + \underbrace{\mathcal{C}(\mathcal{H}, n, \delta)}_{\text{Complexity Price}} + \underbrace{\inf_{f \in \mathcal{H}} R(f) - R(f^*)}_{\text{Approximation}}$$

where:

- $\mathcal{C}(\mathcal{H}, n, \delta) = O\left(\sqrt{\frac{\text{Complexity}(\mathcal{H}) + \log(1/\delta)}{n}}\right)$
- Complexity can be VC, Rademacher, covering numbers, etc.
- The universe charges interest on complexity!

Wheel 8: The Epiphany of Theory

Climbing the Mountain of Generalization



*"We climbed the mountain of theory not to conquer it,
but to see how small our empirical concerns look from the summit."*

- **From Markov to Hoeffding:** Capturing the "exponential blessing."
- **From VC to Rademacher:** Measuring the richness of function spaces.

Wheel 8: The Theoretical Revelation

What We Have Learned in Our Ascent

The Sacred Gifts:

- **Concentration:** Probabilities decay exponentially fast.
- **Uniformity:** Bounds that hold for *all* functions in $\mathcal{H}$.
- **Complexity:** Quantifying the "richness" of your model.

The Eternal Truths:

- **Data size** beats algorithm complexity.
- **Simplicity** is a mathematical proxy for generalization.
- **Learning** is probabilistically guaranteed, not accidental.

The Wheel 8 Benediction

May your models generalize and your bounds be tight!

May your confidence intervals contain the truth!

**For we have seen the mathematics behind the magic,
And found it more beautiful than the magic itself.**

Wheel 9a: Deployment & Impact

Where Learning Machines Meet Reality

The Full Circle Revelation

“We began with a problem in Wheel 1.
We end with a solution in Wheel 9.”

The Deployment Trinity:

- ① **Model:** $\hat{f}_n$ (Wheels 1–8)
- ② **Infra:** Software, hardware, APIs
- ③ **Ops:** Monitoring, drift, impact

The Fundamental Equation:

$$\text{Impact} = \text{Acc} \times \text{Scale} \times \text{Adoption}$$

The Deployment Gap:

- 95% of models never reach production.
- 70% of AI projects fail at deployment.
- Value $\neq$ Test Set Accuracy.

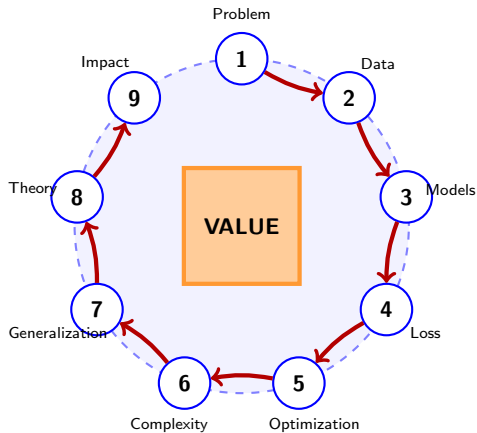
$$\mathbb{I}_{\text{Value}} = \mathbb{I}_{\text{Deployed}} \cdot \mathbb{I}_{\text{Used}} \cdot \mathbb{I}_{\text{Effective}}$$

The Final Test

A model's true test is not its test set accuracy, but its impact on the world it was built to serve.

Wheel 9b: The 9-Wheel Journey

Visualizing the Path from Problem to Solution



"The end of our journey is just the beginning of the model's life."

Wheel 9: The Scalability Challenge

From Notebook to Nation-Scale

Scaling Dimensions:

- **Data:** n from thousands to billions
- **Features:** p from hundreds to millions
- **Requests:** QPS from 10 to 100,000+
- **Latency:** From seconds to milliseconds
- **Cost:** From \$10 to \$10M/month

The Scalability Equation:

$$\text{Throughput} = \frac{\text{Resources}}{\text{Latency} \times \text{Memory}}$$

$$\text{Cost} = \text{Throughput} \times \text{Time} \times \text{Price/Unit}$$

Architecture Patterns:

- **Monolithic:** Single service, simple
- **Microservices:** Decoupled, scalable
- **Serverless:** Event-driven, pay-per-use
- **Edge:** On-device, low latency
- **Federated:** Distributed, privacy-preserving

Deployment Strategies:

- **Shadow:** Run parallel, no decisions
- **Canary:** Gradual rollout
- **Blue-Green:** Instant switchover
- **A/B Testing:** Compare variants

The Scaling Journey: From Prototype to Production

Wheel 9: Monitoring & Maintenance

The Never-Ending Work of Keeping Models Alive

The Harsh Reality

Models degrade. Data drifts. Users change. The world evolves.

$$\mathbb{P}(\text{Model fails in production}) \xrightarrow{t \rightarrow \infty} 1$$

What to Monitor:

- **Input Drift:** $\Delta p(\mathbf{x})$ over time
- **Concept Drift:** $\Delta p(y|\mathbf{x})$ over time
- **Performance:** Accuracy, latency, throughput
- **Business Metrics:** Revenue, retention, satisfaction
- **Fairness:** Performance across subgroups

Drift Detection:

$$D_{\text{KL}}(p_{\text{train}} \| p_{\text{prod}}) > \tau \quad \Rightarrow \quad \text{ALERT!}$$

Retraining Strategies:

- **Scheduled:** Weekly, monthly, quarterly
- **Triggered:** When metrics degrade
- **Continuous:** Online learning
- **Incremental:** Update, don't retrain

The MLOps Lifecycle:

- 1 Development (Wheels 1-8)
- 2 Testing & Validation (Wheel 7)
- 3 Deployment (Wheel 9)

Wheel 9: Ethical Deployment & Responsible AI

The Moral Dimensions of Machine Learning

The Ethical Imperatives:

- **Fairness:** Equal performance across groups
- **Transparency:** Explainable decisions
- **Privacy:** Protect user data
- **Security:** Robust to attacks
- **Accountability:** Clear responsibility

Fairness Metrics:

Demographic Parity : $\mathbb{P}(\hat{Y} = 1|A = a) = \mathbb{P}(\hat{Y} = 1|A = b)$

Equal Opportunity : $\mathbb{P}(\hat{Y} = 1|Y = 1, A = a) = \mathbb{P}(\hat{Y} = 1|Y = 1, A = b)$

Equalized Odds : Both above hold

Explainability Methods:

- **Global:** Feature importance, LIME
- **Local:** LIME, SHAP, counterfactuals
- **Example-based:** Prototypes, analogs
- **Rule-based:** Decision trees, model rules

Privacy Techniques:

- Differential Privacy: (ϵ, δ) -guarantee
- Federated Learning: Data stays on device
- Homomorphic Encryption: Compute on encrypted data
- Secure Multi-party Computation

Fairness
Ideal Balance

Wheel 9: The Return to Wheel 1

Closing the Circle, Starting Anew

The Great Revelation

**Wheel 9 is not an end, but a beginning.
It completes the circle by returning to Wheel 1.**

The Feedback Loop:

- ① **Wheel 1:** Define problem
- ② **Wheels 2-8:** Build solution
- ③ **Wheel 9:** Deploy solution
- ④ **Learn from deployment**
- ⑤ **Return to Wheel 1:** Refine problem

What We Learn in Production:

- Real users Test set
- Real costs Academic metrics
- Real constraints Theoretical assumptions

The Evolution Equation:

$$\text{Model}_{t+1} = \text{Model}_t + \alpha \cdot (\text{Feedback}_t - \text{Expectation}_t)$$

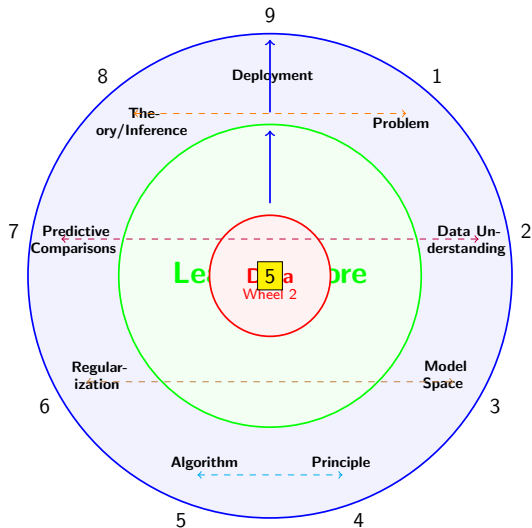
where α is the learning rate from reality.

The Maturity Scale:

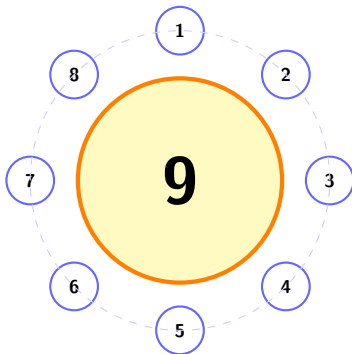
- **Level 1:** One-off analysis
- **Level 2:** Reusable scripts
- **Level 3:** Production pipelines
- **Level 4:** Automated retraining
- **Level 5:** Continuous improvement

The Complete 9-Wheel System

A Unified Theory of Statistical Machine Learning



THE NINE WHEELS TURN



Problem → Data → Model → Principle → Algorithm
→ Regularization → Validation → Theory → Deployment

The Journey Complete

Final Reflections

The Statistician's View

"To understand God's thoughts, one must study statistics, the measure of His purpose."

— *Florence Nightingale*

The Architect's View

"The formulation of a problem is often more essential than its solution."

— *Albert Einstein*

Congratulations!

You have traversed the landscape from empirical risk to theoretical guarantees. Go forth and build models that matter.

Wheel 9: Deployment & Production

From Model to Impact: Software, Scale, and Society

The final, crucial step is moving from a script ($\hat{f}_{\text{script}}$) to a deployed asset ($\hat{f}_{\text{prod}}$).

- **Software & APIs:** Package the model into a reusable component (e.g., a Python class, a Docker container). Expose it as an API (e.g., REST, gRPC) for other applications to consume.
- **Scalability:** The model must handle production load.
 - **Vertical Scaling:** More powerful machines.
 - **Horizontal Scaling:** Replicate the model API across many machines behind a load balancer.
- **Deployment Patterns:**
 - **Shadow Mode:** Run new $\hat{f}_{\text{prod}}$ in parallel with the old system without affecting decisions.
 - **Canary Deployment:** Roll out to a small subset of users first.
 - **A/B Testing:** Compare two models live to measure the impact on business metrics.
- **Monitoring:** Track prediction latency, throughput, and—crucially—**data drift** ($\Delta p(\mathbf{x})$) and **concept drift** ($\Delta p(y|\mathbf{x})$). Trigger $\mathbb{1}_{\{\text{Retrain}\}}$ when performance degrades.

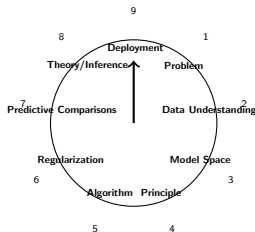
The Complete ML Wheel: A Unified View

From Problem to Production and Beyond

"The whole is greater than the sum of its parts."
- Aristotle

The Interlocking System

The machine only learns when all nine wheels turn together. Neglecting any single wheel compromises the entire statistical learning process.



The Cycle Continues

Model building is not linear but cyclical. Success requires spinning these wheels continuously, with each revolution creating more robust and valuable systems.

The Nine-Wheeled Benediction

Wheel	The Blessing of Learning
1. Problem	May your problems be well-formulated and intentions clear.
2. Data	May your data be representative and your noise be Gaussian.
3. Model	May your hypothesis class $\mathcal{H}$ contain the true f^* .
4. Principle	May your Loss Functions ℓ be convex and your risks bounded.
5. Algorithm	May your gradients be steep and your convergence be certain.
6. Regularization	May your penalties λ find the sweet spot of the tradeoff.
7. Validation	May your CV-error be an honest mirror of reality.
8. Theory	May your Rademacher complexity $\mathfrak{R}_n$ shrink as $1/\sqrt{n}$.
9. Deployment	May your impact be measured in lives improved, not just bits.

*"The model is but a shadow; the data is but a sample.
Only the **generalization** is eternal."*